OPEN ACCESS

Prevention strategies and modifiable risk factors for concussion: a systematic review and meta-analysis for the Female, woman and girl/or Athlete Injury pRevention (FAIR) consensus

Isla J Shill ,[1,2,3] Heather A Shepherd ,[4] Paul H Eliason ,[2] Ash T Kolstad ,[2] Omar Heyward ,[5,6] Géraldine Martens ,[7] Kerry Peek ,[8,9] Clara A Soligon ,[2] Matthew G King ,[10] Stephen W West,[2,11,12] Osman Hassan Ahmed ,[13,14,15] Cheri A Blauwet,[16] Steven Broglio ,[17] Araba Chintoh ,[18,19] Jean-Michel Galarneau ,[2] K Alix Hayden ,[20] Sharief Hendricks ,[5,21] Michael Makdissi ,[22,23] Debbie Palmer ,[1,3] Stacy Sick,[2] Jackie L Whittaker ,[24,25] Kay M Crossley ,[23] Kathryn J Schneider ,[2] Carolyn A Emery [2]

► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/bjsports-2025-109915).

For numbered affiliations see end of article.

Correspondence to
Professor Carolyn A Emery; caemery@ucalgary.ca

Accepted 25 August 2025

Check for updates

© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: Shill IJ, Shepherd HA, Eliason PH, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2025-109915

ABSTRACT

Objective To examine prevention strategies and potential modifiable risk factors (MRFs) for sport-related concussion (SRC) and head impact/head acceleration event (HAE) outcomes in female, woman and/or girl athletes.

Design Systematic review with meta-analyses and Grading of Recommendations, Assessment, Development and Evaluation.

Data sources Medline, CINAHL, PsycINfo, SportDiscus, ERIC, CENTRAL and CDSR.

Eligibility Primary data studies with comparison group(s) assessing the association of prevention interventions and/or MRFs for SRC or HAE with ≥1 female/woman/girl in each study group.

Results Of the 108 included studies, 67 evaluated a SRC prevention strategy (equipment n=25, policy/rule n=21, training n=10, management n=11) and 41 evaluated potential MRFs (34 distinct MRFs across nine categories). In total, 40/108 (37%) studies (prevention 19/67; MRF 21/41) included female/woman/girl-specific estimates. Three meta-analyses were conducted: two SRC prevention strategies (headgear, eyewear) and one MRF (artificial turf vs grass) based on availability of female/woman/girl-only estimates and similar outcomes and exposure. Headgear was associated with 30% lower SRC rates in adolescent female/girl lacrosse and soccer (IRR=0.70, 95% CI 0.50 to 0.99; very-low certainty). Eyewear use was not protective for SRC (IRR=1.08, 95% CI 0.69 to 1.68; very-low certainty). SRC rates did not differ by artificial turf versus grass (IRR=0.95, 95% CI 0.62 to 1.45; very-low certainty).

Conclusion We found limited evidence for prevention strategies and MRFs in female/woman/girl athletes except for very-low certainty evidence supporting headgear use in adolescent lacrosse and soccer. Future studies should consider the design, implementation and evaluation of SRC prevention strategies that target MRFs to guide safe practice recommendations specifically for female/woman/girl athletes.

INTRODUCTION

Sport-related concussions (SRC) are a public health concern with elevated rates for female, woman and/or girl (hereafter female/woman/girl) athletes relative to males/men/boys.[1–3] The higher burden of SRC in female/woman/girl athletes may be related to sex-specific and/or gender-specific physiological, biomechanical and/or sociocultural factors and mechanisms.[4 5] These could be linked to hormonal differences, differences in neck strength/control, a higher likelihood of reporting a concussion, or mechanistic differences, such as heading a ball in soccer or tackle characteristics within rugby.[1 6–8]

SRC primary prevention strategies can include training programmes (eg, neuromuscular training warm-up, tackle training), equipment (eg, headgear, mouthguards) and policy/rule changes (eg, body checking policy, reduced legal tackle height).[9] Secondary prevention strategies facilitate early detection and care following SRC to reduce recurrence.[10 11] Potential modifiable risk factors (MRFs) consider intrinsic (individual) and extrinsic (environmental) factors to inform the development of targeted prevention strategies to reduce SRC incidence rates, severity and burden.[11 12]

The Consensus Statement on Concussion in Sport (Amsterdam 2022) provided recommendations for the prevention, diagnosis and management of SRC and head impact (ie, typically measured through video analysis and expressed as frequencies and/or rates) and/or head acceleration events (HAEs) (ie, typically measured through wearable technologies such as instrumented mouthguards to measure linear and rotational acceleration) (hereafter, head impact/HAE).[10] Only 3% of studies informing prevention and modifiable risk factor recommendations in the Consensus Statement were female/woman/girl focused.[13] Although some recommendations translate, studies reporting aggregate data across males/man/boy and female/woman/girl athlete contexts may not be appropriate. SRC prevention strategy

WHAT IS ALREADY KNOWN

⇒ Female/woman/girl athletes experience high rates of sport-related concussion, particularly in collision sports.
⇒ It is unclear if there are sex-specific or gender-specific injury prevention approaches or risk factors for sport-related concussion.
⇒ Prevention strategies (eg, mouthguards, headgear, policy disallowing body checking in child/adolescent ice hockey, limited contact practice in American football, neuromuscular training in rugby and concussion management) have been shown to reduce concussion rates in primarily male, man and/or boy sports.

WHAT ARE THE NEW FINDINGS

⇒ Only 37% of included studies report female/woman/girl athlete-specific findings that assessed prevention interventions or identified modifiable risk factors for sport-related concussion.
⇒ Very-low certainty evidence (two cohort and one RCT) demonstrated headgear use reduced sport-related concussion rates by 30–39% in adolescent female and/or girl lacrosse and soccer.
⇒ Very-low certainty evidence (three cohort studies) indicated protective eyewear is not protective of concussion in adolescent female/girl lacrosse.
⇒ Very-low certainty evidence (three cohort studies) revealed soccer-related concussion rates do not differ between grass and artificial turf fields for adult female and/or woman players.
⇒ There is limited evidence on training, equipment or policy/rule changes for preventing sport-related concussion in female/woman/girl athletes.
⇒ The absence of female/woman/girl Para sport athlete-specific data highlights the significant need for further research on sport-related concussion prevention strategies and modifiable risk factors in athletes with disabilities.

evaluations specific to female/woman/girl athlete contexts are needed to inform female/woman/girl athlete recommendations. Globally, female/woman/girl participation rates in contact/collision sports (eg, rugby, soccer) are growing.[14–17] There is an urgent need for sex-specific and/or gender-specific concussion and head impact/HAE prevention strategies for female/woman/girl athletes.

The primary research questions addressed by this systematic review were:

1. What prevention strategies (eg, training strategies, policy/rules, personal protective equipment, management) reduce SRCs, recurrent SRCs and/or head impacts/HAEs in female/woman/girl athletes, and are there unintended consequences?
2. What are the potential MRFs for SRCs and/or head impacts and/or HAEs in female/woman/girl athletes?

METHODS

Registration

This review was registered in the PROSPERO registry (CRD42023485808).

Framework

The Cochrane Handbook, Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines and PRISMA-Search extension informed the conduct and reporting of this review (online supplemental material 1).[18–20] Equity, diversity and inclusion, and patient and public engagement statements are provided in Box 1 and Box 2, respectively.

Box 1 Equity, diversity and inclusion statement

In this review, we refer to participants with the terms 'female, woman and/or girl' athletes (with the abbreviation of female/woman/girl) and 'male, man and/or boy' athletes (male/man/boy). We recognise the terms are not synonymous and may mean different things to different people. Also, we use the term 'sex and/or gender' (sex/gender). We acknowledge the active conversation on the meaning and definitions of gender and sex, and note that the term female(s) is not intended to reduce humans to their biological sex, and terms women and girls are not intended to reduce humans to their gender.

The Female, woman and girl Athlete Injury pRevention (FAIR) Consensus Steering Committee planned and designed this review. The recommendations and FAIR Consensus will be developed in consultation with the FAIR External Advisory Committee. This group (n=24, 80% women; n=17, 71%) affiliated with International Olympic Committee (IOC) Research Centres for prevention of injury and protection of athlete health includes academics and sports medicine/health professionals (n=19; 79%) with broad expertise ranging from epidemiology (n=18; 75%), injury prevention (n=21; 88%), sport sciences (n=4; 17%), health promotion (n=2; 88%), and across career stage (n=7 late-career, n=9 mid-career, n=8 early career) from 4 continents, who are mostly white, and from well-resourced countries.

For this systematic review, authors were chosen by the FAIR Steering Committee based on relevant and diverse experiences, backgrounds, expertise and professions. Authors were 58% women (n=14); 79% white (n=19); 4% from middle-to-low income birth countries (n=1); 13% ability limited (n=3), with self-reported lived experience as an athlete (n=11; 46%), coach (n=9; 38%), referee (n=2; 8%), health, medical or exercise practitioner (n=13; 54%), and member for sport organisation (n=8; 33%), government (n=2; 8%), or healthcare (n=7; 29%) committees across paediatric (n=20; 83%), adult (n=17; 71%), and Para sport (n=14; 58%) spanning four continents and career stage (n=3 graduate students; n=9 early-, n=6 mid-, n=6 late-career). The author demographics related to the current concussion prevention and modifiable risk factor review are in online supplemental material 4. We recognise that the limited number of authors who are not white, from low-income countries, and ability-limited may also reflect the limited diversity of the study participants included in studies included in this review.

Data sources and search strategy

Relevant studies were identified by searching eight databases (Medline ALL-OVID, CINAHL Plus with Full Text-EBSCOhost, APA PsycINFO-OVID, Cochrane Database of Systematic Reviews-OVID, CENTRAL-OVID, SPORTDiscus with Full Text-EBSCOhost EMBASE-OVID, ERIC-EBSCOhost). Search strategies consisted of subject headings and text words related to (1) SRC, head impact and HAE; (2) sports and (3) prevention and risk factors, with limits for publication (2001 to current), consistent with the International Consensus Statement on Concussion

Box 2 Patient and public involvement

FAIR Consensus External Advisory Committee (EAC): The FAIR EAC consists of 8 women with lived experiences as elite (n=1 Olympian, n=1 Paralympian) and youth (n=1) athletes, team physicians/physiotherapists (n=3), coaches (n=2), sport scientists (n=1), and leadership roles in sport (n=5), government (n=2), industry (n=1), and healthcare (n=5) organisations spanning multiple ethnicities (62.5% white), birth countries (50% from low to middle-income), and abilities (n=4 Para sport). The EAC level of engagement is consistent with the International Association for Public Participation 'consult' or 'involve' level with tasks including reviewing and providing feedback on FAIR consensus and recommendations. The EAC will also participate in future knowledge translation activities with tasks including reviewing and providing feedback on systematic review summaries and papers, consensus and recommendations.

in Sport (Amsterdam 2022) methodology.[10] The concepts for 'sports' and 'prevention' were adapted from searches conducted for the International Consensus Statement on Concussion in Sport (Amsterdam 2022).[10 13] A health sciences librarian (KAH) developed the MEDLINE search strategy in consultation with the methods (IJS) and lead authors (CAE, KJS), which was pilot tested with six seed studies. KAH then adapted the search strategy for other databases. An external health sciences librarian reviewed the search strategy. KAH performed the searches on 30 October 2023 and 23 November 2023 for CENTRAL and CDSR (online supplemental material 2) and uploaded records to an online screening platform (Covidence, Veritas Health Innovation) to support independent and blinded title/abstract and full-text screening. The Cochrane systematic review database was searched to enable hand-searching of the reference lists from systematic reviews.

Eligibility

Included studies met the following criteria: (1) English language, (2) human participants, (3) original data, (4) outcome of SRC and/or mild traumatic brain injury (mTBI) and/or concussion severity and/or recurrent concussion and/or head impact and/or HAE and/or unintended consequences, (5) specific prevention intervention or MRF, (6) comparison group and (7) at least one female/woman/girl participant in each study group. Unintended consequences include any adverse results that are presented from the evaluation of a concussion prevention strategy (eg, change in injury epidemiology, change in game metrics). Para sport was specifically included to encompass an inclusive sport population. Exclusion criteria included review articles, commentaries, letters-to-editor, case-series, case-studies, reports, abstracts, conference proceedings, dissertations and pre-experimental study designs.

Study selection

After removing duplicates, a three-step screening process was conducted in Covidence: (1) rapid review, (2) title/abstract screen and (3) full-text screen.[21] First, individual reviewers (IJS, PHE, OH, ATK, CAS, SS, CAE) performed a rapid review to remove any clearly irrelevant studies (eg, non-human based studies, commentaries). Remaining title/abstracts were screened by two independent authors (IJS, MGK, CAE, AC, KJS, OHA, SB, PHE, SH, CAS, OH, DP, GM, SWW, MM, ATK, KP, SS). Finally, the full text of remaining records was independently reviewed by CAE, IJS, PHE or ATK and one other reviewer (KJS, OHA, SB, AC, SH, OH, GM, MM, MGK, DP, KP, SS, CAS, SWW, HAS). Discrepancies at title-abstract or full-text screening were resolved by the lead (CAE) or methods (IJS) author. Before screening, inter-rater agreement was established (≥80% agreement) between all raters and a consensus between lead (CAE) and methods author (IJS) based on a sample set of 50 records.

Data extraction

Two authors independently extracted data from studies. Discrepancies were discussed by raters for consensus, or a third reviewer (IJS, CAE) resolved the disagreement. Extracted data included the following: authors, year, country of study, study design, outcome definition (SRC, SRC severity, head impact/HAE), comparison group(s), demographics (sample size, age, sex/gender, sport, sport-level), intervention, MRF and results (descriptors, point estimates, variability). All potentially modifiable risk factors were extracted by two authors, with consensus on disagreements achieved and/or a third author's input to achieve consensus. MRFs were categorised following data extraction to summarise across categories including arthropometric, lifestyle, training, competition, skill/technique, environmental, muscle function and ball characteristic factors.

Risk of bias

Dyads from 18 reviewers (IJS, MGK, CAE, AC, KJS, OHA, SB, PHE, SH, CAS, OH, DP, GM, SWW, MM, ATK, KP, SS, HAS) independently assessed the quality of evidence across studies in duplicate, based on criteria for transparent reporting, internal validity (study design, reporting quality, presence of selection and misclassification bias, potential confounding) and external validity (generalisability) with the 27-item Downs and Black quality assessment tool (DBQAT; online supplemental material 3).[22] A risk-of-bias score was assigned for intervention studies based on 22-items (omitting items 1, 4, 6, 8) and for MRF studies based on 16-items (omitting items 1, 4, 6, 8, 13–15, 17, 19, 23, 24, 27 for non-intervention studies). Disagreements were resolved through consensus and/or a third reviewer (IJS, HAS). The percentage of items at low risk of bias (achieved maximum rating) was calculated for each study.

Data syntheses

Prevention interventions and potential MRFs were categorised by type, and female/woman/girl-specific data were disaggregated. Relative risk (RR; 95% CI), incidence rate ratio (IRR; 95% CI), OR (95% CI) or HR (95% CI) for interventions or MRF was summarised and calculated where data were available.

Meta-analyses or semi-quantitative syntheses were completed when ≥3 included studies reporting female/woman/girl with similar intervention or MRF (and outcomes) were identified with disaggregated female/woman/girl data reported. Intent-to-treat and per-protocol (as-treated) analyses were included as applicable. Meta-analyses considered random effects models to account for between-study heterogeneity and used a restricted maximum likelihood model. Stata V.18 was used to conduct all meta-analyses.[23]

Certainty of evidence for interventions and MRFs was rated by three authors (IJS, KJS, CAE) using Grading of Recommendations Assessment, Development, and Evaluation (GRADE).[24] Potential MRFs were assessed using a modified approach considering study design strength.[24] If the criteria for meta-analyses were not met, the certainty of evidence was determined using the same GRADE or modified GRADE criteria adapted for non-pooled data. A rating based on domains, a statement of certainty

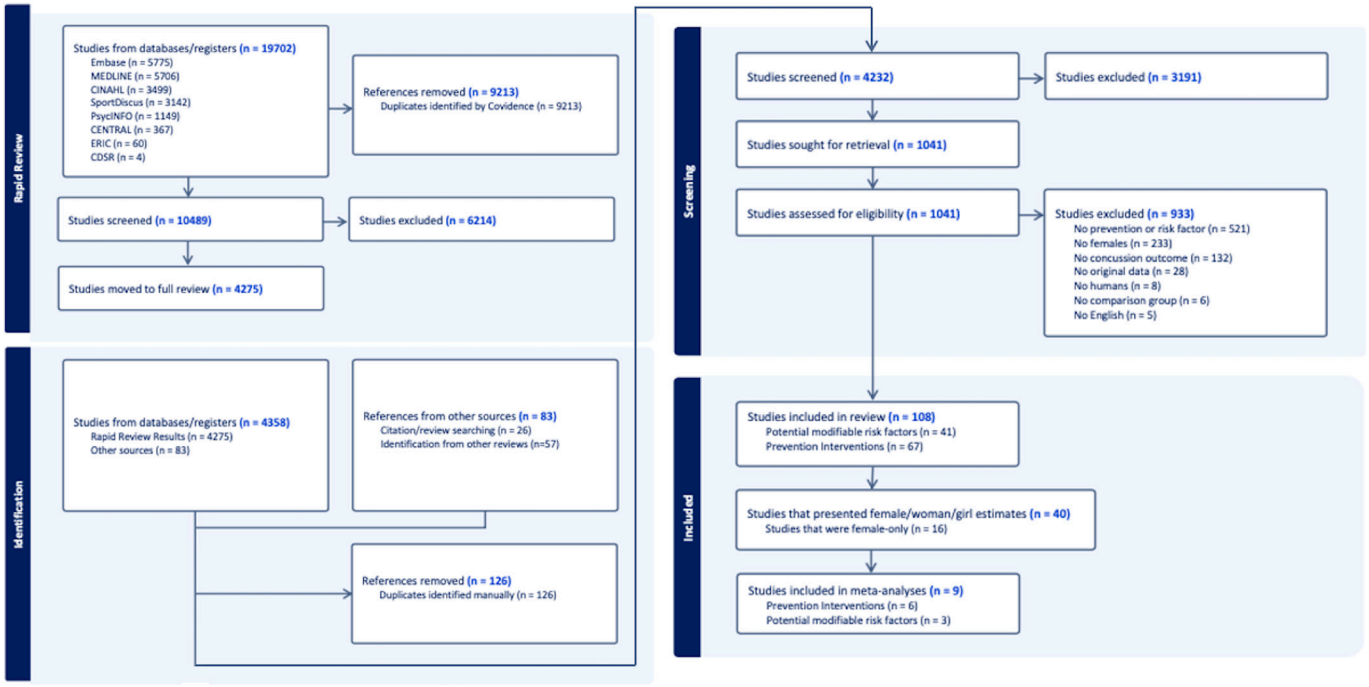


Figure 1 Study identification Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

based on consistency and directionality, and magnitude of effect was completed by lead/methods authors (IJS, KJS, CAE). Semi-quantitative GRADE approach for pooled and non-pooled data is described in online supplemental material 4.

RESULTS

In total, 108 studies (67 prevention, 41 MRFs) were included (figure 1). 40 (37%) reported female/woman/girl-specific estimates and 72 (67%) included participants under age 18. 27 (25%) studies reported girl (paediatric ages ≤18) specific data. No studies reported female/woman/girl Para athlete-specific data.

Study characteristics

The 108 studies included 7 randomised controlled trials, 5 quasi-experimental, 67 cohort, 7 case-control, 21 cross-sectional and 1 case-cross over study representing 12 countries and 13 individual sports. Across all studies, five outcomes of interest were assessed (SRC: 74 studies; HAE: 20 studies; SRC severity: 15 studies; head impact: 8 studies; recurrent SRC: 4 studies). Online supplemental material 5 shows aggregate and disaggregate female/woman/girl and male/man/boy study results for all included studies.

Prevention strategy evaluation

Prevention strategies spanned training (15%; 10/67), equipment (37%; 25/67), policy/rule (31%; 21/67) and management (16%; 11/67) interventions. Table 1 displays the female/woman/girl results of the 19 (28%) prevention strategy studies across seven sports that presented female/woman/girl-specific estimates (most common: soccer (37%; 7/19); lacrosse (21%; 4/19)). Nine of the studies that provided female/woman/girl-specific estimates were female/woman/girl athlete-only evaluations. After considering the studies investigating relationships between a unique intervention and SRC, mTBI, head impact or HAE, plus availability of female/woman/girl-specific estimates and estimate type (eg, IRR, OR, HR) reported, a meta-analysis and semi-quantitative analysis were performed for two interventions (headgear, eyewear). No semi-quantitative analyses were performed for non-pooled data due to heterogeneity of outcomes and interventions. In studies that reported female/woman/girl-specific estimates, none examined unintended consequences.

Training

Training programme strategies[25–34] were examined in 10 (15%; 10/67) prevention strategy studies (four training studies (40%; 4/10) included female/woman/girl-specific estimates;[27 30 32 33] two (20%; 2/10) were female/woman/girl-only[32 33]). Considering studies that contained female/woman/girl-specific estimates, all were performed in soccer.[27 30 32 33] There were conflicting results from studies assessing neck strength training (FIFA 11+ with added neck strengthening exercises),[27 30] and heading training (neck and core strengthening) programmes in female/woman/girl soccer.[32 33] One RCT evaluating neck strength training found no association between neck strength training and changes in HAE magnitude (head kinematics),[27] while another RCT demonstrated 27.7% ($F_{(1,40)}=2.68$, $\eta^2=0.21$ (0.03, 0.36)) lower peak angular head velocity in female/girls completing the FIFA 11+ with added neck exercises, compared with the standard FIFA 11+.[30] No training-related effects were observed on head impact kinematics in two quasi-experimental studies after a 3–5-month heading training programme ($p>0.05$) including neck and core strengthening in adolescent female/girl soccer.[32 33]

Policy/rule change

Policy/rule change strategies were evaluated in 21 (21/67; 31%) studies,[35–55] where four policy/rule change cohort studies (4/21; 19%) included female/woman/girl-specific estimates.[47 49 50 53] These four studies evaluated head-kick regulations in taekwondo,[47] a heading ban in soccer,[49] a change in head kicking rule in karate[50] and the implementation of fair play rules in ice hockey.[53]

A 2013 sparring-taekwondo competition rule change acknowledging simple light touch head kicks (rather than

Table 1 Female/women/girl results of prevention intervention studies with female/woman/girl estimates

Author year Country; study design FWG* Paediatric†	Type of prev	Participant characteristics sport sample characteristics	Definition of outcome (concussion, head impact)	Intervention (IG) and control (CG) group descriptions	Effect estimates (eg, incidence rate ratios (IRR), OR) reported or calculated based on data provided (95% CI where applicable) or other relevant main findings	Downs and Black Score
Mansell 2005 USA; RCT FWG: 53%‡ Paediatric: N	Training	Soccer; National Collegiate Athletic Association Division 1 Collegiate age=19.16 ±0.898 n female=19	Head acceleration (Peak head-neck segment angular acceleration).	IG: 8 weeks cervical resistance training programme twice a week (+ offseason weightlifting/ conditioning sessions 2 days/week). CG: offseason weightlifting/ conditioning sessions 2 days/week.	No significant differences before and after the training intervention in flexion or extension movements.	19
Peek 2022 Australia; RCT FWG: 40%‡ Paediatric: Y	Training	Soccer; National Premier League 12–17 years n=21 players (IG: N=2 female teams, n=14 female; CG: N=2 female teams, n=7 female)	Head acceleration (PLA, PAV).	IG: FIFA 11+with additional neck exercises (3×/week for 5 weeks). CG: FIFA 11+ without additional neck exercises (3×/week for 5 weeks).	PLA: F (1,40)=3.25, effect size h^2=0.14 (95% CI 0.01 to 0.28) PAV: F (1,40)=2.68, effect size h^2=0.21 (95% CI 0.03 to 0.36); PAV reduced 27.7% IG versus CG.	19
Wahlquist 2021 USA; Quasi-exp FWG: 100% Paediatric: Y	Training	Soccer Under-12 All female; age=mean 10.5 years (SD 0.5) n=12 (attrition 25) CG=pre-season, IG=post-season	Head acceleration (PLA, PRV, PRA).	IG: Get aHEAD safely in Football programme: 1×/week header drill(s); 2×/week neck/core strengthening exercises. CG: No intervention.	No changes in PLA, PRA, PRV. No statistics reported.	16
Wahlquist 2022 USA; Quasi-exp FWG: 100% Paediatric: Y	Training	Soccer Under-12 All female n=2, n=27 (IG: N=1 team; n=14 (age=mean 10.6 SD 0.5); CG: N=1 team; n=13 (age=mean 11.0 SD 0.4))	Head acceleration (PLA, PRV, PRA).	IG: Get aHEAD safely in Football programme: 1×/ week: header drill(s); 2×/week neck/core strengthening exercises. CG: No intervention programme.	No significant differences between IG and CG. PLA: p=0.692; PRA: p=0.379; PRV: p=0.852	14
Jae-Ok 2016 South Korea; Cohort FWG: 50%‡ Paediatric: N	Policy/law change	Taekwondo World Championship competition n female=32 IG: n female=16; CG: n female=16	Head contact (head kick).	IG: Definition of head kick expanded to a soft touch. CG: Head kicks were included if they generated force.	RR: /1000 athlete-exposure: 2.0 (95% CI 0.69 to 6.49), p=0.17 /1000 match-exposure: 2.09 (95% CI 0.72 to 6.78), p=0.14	6
Lalji 2020 USA; Cohort FWG: 39%‡ Paediatric: Y	Policy/law change	Soccer 10–13 years IG: n female=1400; CG: n female=1520	Concussion.	IG: Heading ban. CG: No heading ban.	Unadjusted OR: 1.31 (95% CI 1.01 to 1.69)	12
Macan 2006 Canada; Cohort FWG: 32%‡ Paediatric: Y	Policy/law change	Karate; National championships, international tournament, Croatia Cup IG: n female=498 matches; CG: n female=287 matches	Concussion.	IG: World Karate Federation rule implementation introduced. CG: Before World Karate Federation rule implementation.	Head injury IRR (95% CI): All: 0.52 (0.34 to 0.79); <18: 0.22 (0.12 to 0.39); ≥18: 0.90 (0.49 to 1.67)	13
Smith 2016 USA; Cohort FWG: 22%‡ Paediatric: Y	Policy/law change	Ice hockey; All levels 11–14 years n female=328 (Under-14A/B) IG: n female=171; CG: n female=157	Concussion, head contact (HHWDC: impact without diagnosed concussion).	IG: Implementation of fair play rules within hockey tournament. CG: No implementation of fair play rules within hockey tournament.	IRR (95% CI) Concussions IG/CG: 2.38 (0.10 to 50.00); p=0.596 HHWDC IC/CG: 0.79 (0.05 to 12.5); p=0.869	16

Continued

Table 1 Continued

Author year Country; study design FWG* Paediatric†	Type of prev	Participant characteristics sport sample characteristics	Definition of outcome (concussion, head impact)	Intervention (IG) and control (CG) group descriptions	Effect estimates (eg, incidence rate ratios (IRR), OR) reported or calculated based on data provided (95% CI where applicable) or other relevant main findings	Downs and Black Score
Baron 2020 USA; Cohort FWG: 100% Paediatric: Y	PPE	Lacrosse; High school All female; Age NR CG: 683 183 athlete-exposure (n=unknown) IG: 22 397 athlete-exposures (n=1585)	Concussion.	IG: F3137 Headgear. CG: No headgear.	IRR all=0.238 (95% CI 0.029 to 0.870) IRR (game)=0.152 (95% CI 0.004 to 0.860) IRR (Practice)=0.431 (95% CI 0.011 to 2.48)	18
Caswell 2020 USA; Cohort FWG: 100% Paediatric: Y	PPE	Field lacrosse; High school varsity All female; Mean age=16.2±1.2 years N=1 team, n=49 players	Head acceleration (PLA, PRV), Head impacts (verified impacts where contact was to player's head).	IG: Players wore Women's LX headgear (Cascade Lacrosse; 2017 season). CG: Players did not wear headgear (2016 season).	Calculated IRR: IRR=1.03 (95% CI 0.39 to 2.71)	15
Herman 2022 USA; Cohort FWG: 100% Paediatric: Y	PPE	Lacrosse; High school All female n=289 school-seasons, n NR Total athlete-exposures: 357 225 (Game: 98 858, Practice: 258 367) IG: n=96 school-seasons, Total athlete-exposures: 357 225 (Game: 98 858, Practice: 258 367) CG: N=193 school-seasons, Total athlete-exposures: 91 074 (Game: 25 733, Practice: 65 341)	Concussion, Concussion severity (# days RTP).	IG: Mandating use of headgear (mandated in Florida). CG: No headgear mandate (other states).	Overall: RR: 0.63 (95% CI 0.41 to 00.97 Games only: RR: 0.57 (95% CI 0.33 to 1.00) Practices only: RR: 0.70 (95% CI 0.35 to 1.41) The significance of the point estimates are assumed based on non-overlapping CI	11
Kriz 2012 USA; Cohort FWG: 100% Paediatric: Y	PPE	Field Hockey; High school All female; Age range: 14–18 N=180 schools (IG=39, CG=141) N=110 submitted data (39% attrition) n NR; Total=329 601 athlete-exposure (IG: 66 286, CG: 263 315)	Concussion.	IG: Mandate on the use of protective eyewear. CG: No mandate on the use of protective eyewear.	IRR=1.04 (95% CI 0.63 to 1.75) p=0.857	10
Kriz 2015 USA; Cohort FWG: 100% Paediatric: Y	PPE	Field hockey, High school All female Age range: 14–18 N=206 schools (IG=117, CG=180) n NR; Total=624 803 athlete-exposure (IG:361 488, CG:263 315)	Concussion.	IG: Mandate on the use of protective eyewear. CG: No mandate on the use of protective eyewear.	IRR=0.77 (95% CI 0.58 to 1.02) p=0.068	10
Lincoln 2012 USA; Cohort FWG: 100% Paediatric: Y	PPE	Lacrosse; High school All female N=25 schools; n NR (IG: 5566 player-seasons; CG: 3864 player-seasons)	Concussion.	IG: Mandated use of protective eyewear. CG: Prior to mandated use of protective eyewear.	IRR=1.6 (95% CI 1.1 to 2.3)	10
McGuine 2020 USA; RCT FWG: 67%‡ Paediatric: Y	PPE	Soccer Age 15.6±1.2 IG: N female=44; n female=1031 CG: N female=44; n female=999 n=17 females dropped out prior to regular season start.	Concussion.	IG: Wearing headgear (Full90 Sports, Premier; Forcefield, Ultra Forcefield Sweatband; LDR Headgear, Soccer Headband; Storelli ExoShield; Unequal Technologies Halo 10 mm). CG: Not wearing headgear.	Intent to treat: RR=0.90 (95% CI 0.55 to 1.48); HR=0.86 (95% CI 0.54 to 1.36) As-treated: RR=0.64 (95% CI 0.38 to 1.08); HR=0.70 (95% CI 0.43 to 1.15) Models consisted of multivariate modelling controlling for covariates (age, sex, year, concussion in last year, symptom severity score) and cluster effect (school)	21

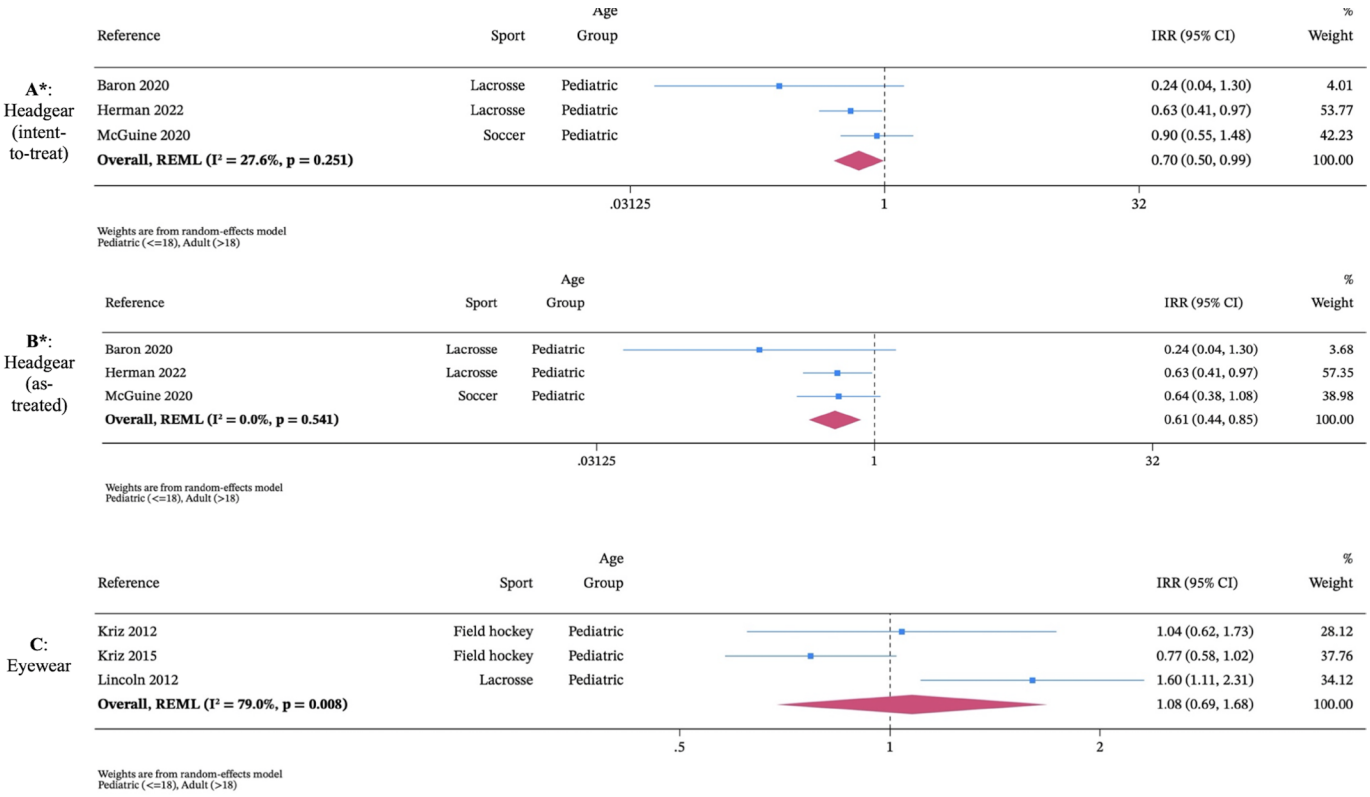
Continued

Table 1 Continued

Author year Country; study design FWG* Paediatric†	Type of prev	Participant characteristics sport sample characteristics	Definition of outcome (concussion, head impact)	Intervention (IG) and control (CG) group descriptions	Effect estimates (eg, incidence rate ratios (IRR), OR) reported or calculated based on data provided (95% CI where applicable) or other relevant main findings	Downs and Black Score
Tierney 2008 USA; Cross-sectional FWG: 66%‡ Paediatric: N	PPE	Soccer; University n women=29 age=19.5 (SD 1.8)	Severity (head injury criteria (HIC)).	IG1: Wearing headband-style headgear. IG2: Wearing cap-style headgear. CG: No headgear.	HIC: No interaction between sex and headgear for HIC (p=0.069, power=0.532). Note: HAE was explored, but only for sex comparisons. No female-only comparisons between IGs and CG were completed.	13
van Ierssel 2021 Canada; Cohort FWG: 27%‡ Paediatric: Y	PPE	Multiple sport IG: n female=80 CG: n female=193	Concussion severity (Post Concussion Symptom Inventory (PCSI)).	IG: Wearing a mouthguard. CG: Not wearing a mouthguard.	PSCI change score of wearing a mouthguard on symptoms at baseline and weeks 1, 2 and 4 (Estimate (95% CI); p value): Baseline: 0.22 (−0.04 to 0.48); p=0.098 Week 1: 0.29 (0.01 to 0.56); p=0.039 Week 2: 0.22 (−0.06 to 0.51); p=0.128 Week 4: 0.08 (−0.20 to 0.36); p=0.581 Models were adjusted for covariates of interest; see online supplemental material 5) for complete list	15
Willmott 2022 Australia; case cross-over FWG: 62%‡ Paediatric: Y	PPE	Australian Football; Under-12, Under-14 Under-14 female (Div 1 N=1 team; Div 2 N=1 team) n female=25 age (mean, SD): 13.04 (1.14)	Head contact (physical contacts to the head/neck (count))	IG: Use headgear. CG: Do not use headgear.	Frequency of head/neck contact events/player: IG versus CG: p=0.896	14
Pierpoint 2018 USA; Cohort FWG: 100% Paediatric: Y	Manage-ment	Soccer, basketball; High school All female IG: N=100 schools; CG: N=36 schools Total Athletic exposures: IG: 1 082 985; CG: 126 266	Concussion.	IG: Athletic trainers in schools. CG: No athletic trainers in schools.	Soccer RR (95% CI): 8.05 (2.00 to 32.51) Basketball RR (95% CI): 4.50 (1.42 to 14.16)	11

*FWG: proportion of sample that is female/woman/girl.
†Paediatric: Inclusive of paediatric (≤18 years old) population yes/no.
‡Studies contain aggregated female/woman/girl and male/man/boy data and/or male/man/boy-specific estimates. These results are presented in Supplemental Material 3, where results of all included studies are reported.
FIFA, Fédération Internationale de Football Association; NR, not reported; PAV, peak angular velocity; PLA, peak linear acceleration; PRA, peak rotational acceleration; PRV, peak rotational velocity; RCT, randomised controlled trial.

requiring forceful contact, 2011 rule) did not change head contact rates across female/woman/girl weight divisions (ages >15; IRR=2.2, 95% CI 0.69 to 6.49).[47] In contrast, the 2000 World Karate Federation rule changes (ie, updated scoring system, stricter rules against prohibited behaviour, and standardising the size and shape of protective gloves) contributed to lower SRC rates for females/girls (ages <18; IRR=0.22, 95% CI 0.12 to 0.39), but not for females/women (ages ≥18; IRR=0.90, 95% CI 0.49 to 1.67).[50] Similarly, implementation of fair play programmes (ie, extra point awarded to team standings for staying below a predetermined number of game-penalty minutes) in female/girl under-14 ice hockey tournaments was not associated with differences in SRC (RR=2.38, 95% CI 0.10 to 50.00) or head contact rates (RR=0.79, 95% CI 0.05 to 12.50).[53] Interestingly, there was an increase in SRC relative to other injuries among female/girl soccer players (unadjusted OR=1.31, 95% CI 1.01 to 1.69) after the US Soccer Federation banned heading for players ages <10, adopted heading restrictions during practice for players ages 11–13, and required concussion education and management improvements.[49]



A*: Intent-to-treat combined match and training **headgear** meta-analysis with concussion outcome and athlete-exposure exposure time for girls, women, and/or females. B*: As-treated combined match and training **headgear** meta-analysis with concussion outcome and athlete-exposure exposure time for girls, women, and/or females. C: Combined match and training **eyewear** meta-analysis with concussion outcome and athlete-exposure exposure time for girls, woman, and/or females.
* One study (McGuine 2020) reported intent-to-treat and as-treated analyses. Both analyses were considered for meta-analyses and ran as a sensitivity analysis.

Figure 2 Meta-analyses of prevention strategies when ≥3 studies when with similar intervention or MRF and outcomes.

Personal protective equipment (PPE)

PPE use was evaluated in 25/67 (37%) studies.[56–80] 10 studies (10/25 (40%) included female/woman/girl-specific estimates;[57 59 64 67–69 73 77 78 80] six (6/25 (24%)) were female/woman/girl-only evaluations.[57 59 64 67–69]

Headgear use

There was very low certainty of evidence in adolescent lacrosse and soccer that headgear for female/woman/girl athletes may reduce SRC IR by 30–39% (intent-to-treat IRR=0.70; 95% CI 0.50 to 0.99; I^2=27.6%; p=0.251 (representing heterogeneity that may not be important); as-treated IRR=0.61; 95% CI 0.44 to 0.85; I^2=0.0%; p=0.541 (representing heterogeneity that may not be important)) (figure 2).[57 64 73] 6 of the 10 female/women/girl PPE studies (60%) explored the effectiveness of headgear in reducing SRC (two cohort, 1 RCT), HAEs (1 cohort, 1 RCT) and head contacts (1 case cross-over study) for females/women/girls.[57 59 64 73 77 80] There are limited data to support a relationship between headgear (in unhelmeted sports) and head acceleration events.

Protective eyewear use

There was very low certainty of evidence of no significant relationship between protective eyewear use and SRC rates (IRR=1.08; 95% CI 0.69 to 1.68; I^2=79.0%, p=0.008 (representing considerable heterogeneity)), based on data from three cohort studies from female/girl-only adolescent field hockey and lacrosse (figure 2; table 2).[67–69]

Mouthguard use

10 studies examined mouthguards,[38 56 58 60 66 74–76 78 79] with 10–25% female/woman/girl athletes per study, and only 1 of the 10 studies (10%), a cohort study, included female/woman/girl-specific estimates.[78] At the time of SRC, females/girls who wore a mouthguard (across multiple sports) reported significantly higher symptom scores 1 week post-SRC compared with non-wearers; however, symptom scores did not differ at 48 hours, 2, 3 or 4 weeks post-SRC.[78]

Management (secondary prevention)

Management strategies were evaluated in 11 (11/67; 16%) studies.[81–91] Three studies (27%; η^2 3/11) examined the use of healthcare personnel,[84 86 88] four studies (4/11; 36%) examined law or policy,[81 87 89 90] and four studies (4/11; 36%) examined time to return to play (RTP).[82 83 85 91] One cohort study (9%; 1/11) included female/woman/girl-specific estimates, reporting that schools with athletic trainers reported higher SRC rates than those without for soccer (RR=8.05, 95% CI 2.00 to 32.51) and basketball (RR=4.50, 95% CI 1.42 to 14.16).[88]

Modifiable risk factors (MRF)

In total, 41/108 (38%) studies evaluated MRFs,[6–8 92–129] with 21/41 (51%) presenting female/woman/girl-specific estimates.[6–8 93 95 98 100 103 105 107–111 113 115–117 122 128 129] Of those 21 studies, 7/41 (17%) were female/woman/girl-only evaluations.[7 8 107 109 110 115 116] Soccer was the most common sport evaluated in MRF studies, identified in 20 (49%;

Table 2 GRADE summary for pooled estimates of prevention strategies and modifiable risk factors for female/woman/girl sport-related concussion

Prevention strategy	Injury type	# of Studies	Participants/ athlete-exposures*	Phase of investigation	Study limitations	Inconsistency	Indirectness	Imprecision	Publication bias	Upgrading factors	Modified GRADE rating	Confidence Statement
Headgear	Concussion	3	2030 participants 1 062 805 athlete-exposures	Phase 2 (2 studies) Phase 3 (1 study)	XX	X	✓	✓	✓	X	+	Minimal confidence headgear use is associated with **lower** SRC rates
Eyewear	Concussion	3	9430 participants 954 404 athlete-exposures	Phase 2 (3 studies)	XX	X	✓	X	✓	X	+	Minimal confidence of an **unclear** association between eyewear use and concussion
Risk Factor	**Injury Type**	**# of Studies**		**Phase of Investigation**	**Study Limitations**	**Inconsistency**	**Indirectness**	**Imprecision**	**Publication Bias**	**Upgrading Factors**	**Modified GRADE Rating**	**Confidence Statement**
Field type	Concussion	3	154 participants 2682 team-matches 797 matches	Phase 2 (3 studies)	X	X	✓	X	✓	X	+	Minimal confidence of an **unclear** association between artificial turf and grass playing fields

X serious limitations, XX very serious limitations, ✓ no limitations, ++++ (high quality), +++ (moderate quality), ++(low quality), + (very low).
*Not all studies reported participant numbers; studies reported participants, athlete-exposures, team-matches, matches.

Table 3 Risk factor summary table

	Intrinsic		Extrinsic	
	Girl, woman, and/or female-specific estimates	*Aggregate*	*Girl, woman, and/or female-specific estimates*	*Aggregate*
Anthropometric properties	Body mass[113]	Body mass index[124] Neck girth/circumference[97 106 120]	None identified	None identified
Lifestyle factors	Hormonal contraceptive use[109] Narcotic use[100]	ADHD medication[92] Alcohol use[101 123] Cigarette use[101 123] E-vape use[123] Recreational marijuana use[101] NSAID use[127] Physical inactivity[123] Sleep[118 121] Sports team participation[123] Steroid use[125]	None identified	None identified
Training factors	Sport specialisation[116 129]	Sport specialisation[96]	Level of contact during scrimmage[128]	None identified
Competition factors	None identified	None identified	Karate competition categories with/without weight bands[93] Continued participation after SRC event[103] Soccer heading game scenarios[7 111]	Karate competition categories with/without weight bands[94] Mixed martial arts bout length[99] Referee experience[99] Use of matchmaker[99]
Skill & technique	Heading technique[107] Tackle characteristics[8] Preparedness and body position into contact[122]	Blocking skill used[112] Head-to-torso range of motion and torso range of motion[97]	None identified	None identified
Environmental factors	None identified	None identified	Surface type Field type[105 108 115] Flooring[128]	Altitude[114 126]
Muscle function characteristics	Neck muscle strength[95 110]	General strength[119] Neck muscle strength[6 97 102 104 120 162]	Not identified	None identified
Ball characteristics	None identified	None identified	Ball size, mass, pressure[117]	None identified
Coaching factors	None identified	None identified	None identified	Highest level of education for coaches[124]

ADHD, attention deficit hyperactivity disorder; NSAID, non-steroidal anti-inflammatory drugs.

20/41) MRF studies. After considering the studies investigating relationships between a unique MRF and SRC, mTBI, head impact or HAE, plus availability of female/woman/girl-specific estimates and estimate type (eg, IRR, OR, HR) reported, a meta-analysis and semi-quantitative analysis was performed for one MRF (field type: natural grass vs artificial turf). No semi-quantitative analyses were performed for non-pooled data due to heterogeneity of outcomes and MRFs.

MRFs included anthropometric properties, lifestyle, training, coaching, competition, skill and technique characteristics, environment, muscle function and ball characteristics. All categories included female/woman/girl estimates except coaching factors. Risk factors are summarised by intrinsic and extrinsic factors for studies that reported female/woman/girl-specific estimates and aggregated female/woman/girl and male/man/boy estimates (table 3).

Anthropometric properties

Five studies examined anthropometric measurements, including body mass, BMI and neck size.[97 106 113 120 124] Only one study (cohort) included female/woman/girl-specific estimates, where a higher body mass was associated with a higher risk of experiencing prolonged SRC symptoms compared with lower body mass among ice hockey athletes (OR=1.07, 95% CI 1.00 to 1.14).[113]

Lifestyle factors

Nine studies examined lifestyle MRFs (eg, hormonal contraceptives, narcotics, ADHD medication).[92 100 101 109 118 121 123 125 127] One study (cohort) evaluated hormonal contraceptive use and one cross-sectional study evaluated narcotics use, included female/woman/girl-specific estimates.[100 109] There was no difference in time to recovery from SRC between females/women/girls who did or did not take hormonal contraceptives ($p>0.05$).[109] Higher symptom severity scores were reported for females/women/girls who had sustained a SRC and did not take hormonal contraceptives compared with those taking hormonal contraceptives ($F_{(1,47)}=5.142$, $p<0.05$).[109] Considering narcotics use, females/girls who participated in softball (OR=3.52 ($p<0.05$)), soccer (OR=3.10 ($p<0.05$)), track (OR=3.21 ($p<0.05$)) or cheerleading (OR=3.55 ($p<0.05$)) had greater odds of one or more previous concussion when they had previously used narcotics one or two times.[100] Females/girls who participated in basketball ($OR_{1\text{-}2x}=3.80$ ($p<0.05$); $OR_{\geq 3x}=3.02$ ($p<0.05$)) or volleyball ($OR_{1\text{-}2x}=3.89$ ($p<0.05$); $OR_{\geq 3x}=2.87$ ($p<0.05$)) had greater odds of SRC history when they had previously used narcotics one or more times.[100]

Training factors

Four studies (cross-sectional) reporting female/woman/girl-specific estimates[116 128 129] included sport specialisation[96 116 129]

and contact level in scrimmage[128] as potential MRFs. In a multisport context (aggregate data), there were conflicting results regarding the association between sport specialisation and SRC with two studies demonstrating no association and one reporting higher odds of concussion in adolescents specialised (≥4/6-point scale) vs nonspecialized (≤3/6) (OR=4.74, 95% CI 1.92 to 11.71; p<0.01).[116] In roller derby, full-contact scrimmage was not associated with SRC (cohort 90% females/women/girls).[128]

Competition factors

Six studies examined MRF related to competition, including type of weight categories,[93 94] order of matches,[99] ongoing sport participation following a suspected SRC,[103] and game scenarios for soccer heading.[7 111] Four studies (three cohort, one cross-sectional) included female/woman/girl-specific estimates.[7 93 103 111] In karate, no significant differences were observed for females/women/girls for SRC when fights occurred in a strict weight class compared with a free-weight team competition.[93] In an evaluation with multiple sports, female/woman/girl athletes continuing to participate in sport following a suspected SRC were more likely to report more severe affective (R^2=0.06, p=0.006) and cognitive-ocular symptoms (R^2=0.03, p=0.02) than those who stopped participating.[103] Within elite female/woman/girl soccer players, significant differences were reported for linear and rotational HAEs for headers resulting from different game scenarios.[7 111]

Skill and technique characteristics

Five studies examined technical characteristics in relation to SRC.[7 8 107 112 122] Three studies (soccer heading technique (one cross-sectional), lacrosse body preparedness before contact (one cohort), rugby league tackle characteristics (one case-control)) included female/woman/girl-specific estimates.[8 107 122] In female/woman/girl soccer, better heading technique was associated with decreased peak head kinematics, lower rotational HAEs during games and increased rotational head acceleration events during practices.[107] In women's lacrosse, no interaction was observed between HAEs and preparedness/anticipation of impact (ie, good body position).[122] Within women's rugby league, head impacts were associated with body positioning into the tackle, location of body contact and proximity location of head to the opposing player.[8]

Environmental factors

Six studies examined environmental factors,[105 108 114 115 126 128] with four (three cohort, one cross-sectional) including female/woman/girl-specific estimates and one of the four a female/woman/girl-only evaluation.[105 108 115 128]

There was very low certainty of evidence of no significant difference in SRC rates based on field type (artificial turf vs grass) used for adult female/woman soccer players (IRR=0.95; 95% CI 0.62 to 1.45; I^2=0.0%, p=0.985) from three cohort studies (figure 3; table 2).[105 108 115] Furthermore, one study with women, men, non-binary/other and trans roller derby athletes reported no association between floor surface type and SRC.[128]

Muscle function characteristics

Nine studies evaluated muscle function.[6 95 97 98 102 104 110 119 120] Eight of these[6 95 97 98 102 104 110 120] explored neck muscle strength or endurance as a MRF for SRC (n=2)[95 98] and head impact magnitude (acceleration and/or velocity) (n=6; including two during soccer heading).[6 97 102 104 110 120] One study explored general strength.[119]

Four studies (three cohort, one cross-sectional) reported female/woman/girl-specific estimates.[6 95 98 110] In one cohort study (lacrosse, soccer, basketball), neck strength was not associated with SRC.[98] In another study, there was no difference in deep neck flexor endurance between female/woman/girl athletes (hockey, soccer, basketball) who did or did not sustain an SRC.[95] Two (one cohort (rugby), one cross-sectional (soccer)) studies explored neck muscle function and head impact magnitude and included female/woman/girl-specific estimates and one of the two studies reported a negative correlational association.[6 110]

Ball characteristics

One cross-sectional study in female/girl soccer players examined ball characteristics (size, mass, pressure) during heading and reported lower ball pressure was significantly related to lower peak linear head acceleration (r^2=0.38, F=103.55, p<0.001) and lower ball mass was significantly associated with lower angular head velocity (r^2=0.56, F=217.93, p<0.001).[117]

Risk of bias

Individual risk-of-bias scores are reported in online supplemental material 6. For studies that reported female/woman/girl estimates, a median DBQAT score of 14 (range 6–21) for prevention strategies and 12 (range 7–18) for risk factor studies was observed. Substantial heterogeneity existed in prevention strategy type and SRC, head impact and HAE outcomes for female/woman/girl studies. Confounding variables were controlled for to some degree in 43% of female/woman/girl studies.

Protocol deviations

There were protocol deviations to the study quality and risk-of-bias instrument. The protocol included the use of the National Institutes of Health suite of Study Quality Assessment tools.[130] To ensure rating consistency across the broad range of study designs included, the DBQAT was used instead.[22] The DBQAT can be used across multiple study designs and includes items that

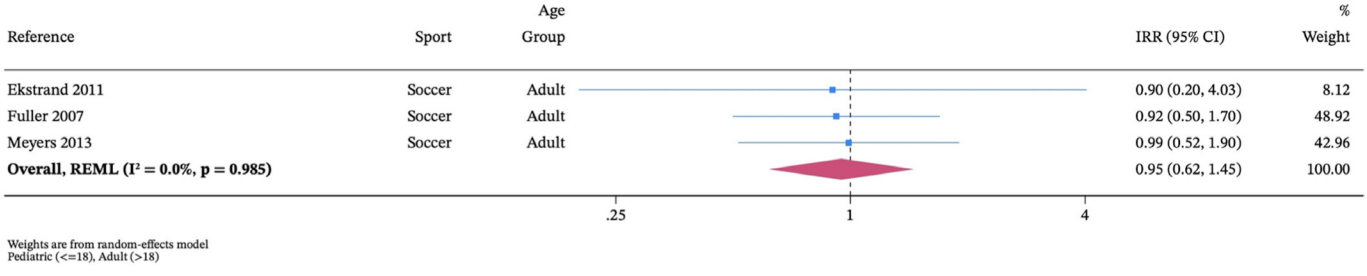


Figure 3 Match field type (artificial turf vs grass) meta-analysis with concussion outcome and player-hours exposure time for girls, women and/or females.

rate internal validity (study design, selection and misclassification bias, potential confounding), external validity (generalisability), and reporting methods.

DISCUSSION

This review examined the current evidence for SRC and head impact/HAE prevention strategies and MRFs in female/woman/girl athletes. The evidence for SRC prevention for female/woman/girl athletes is limited, and heterogeneity exists across prevention strategies, MRFs and outcomes of interest.

Training

There is inconsistent evidence about the efficacy of training strategies to reduce female/women/girl SRCs and head impacts/HAEs.[27 30 32 33] These findings align with previous systematic reviews' aggregated sex/gender data.[13] Variations in training programme components, sports assessed and/or player positions may help to explain the inconsistencies across studies.[13] For example, all training programme strategies studies with female/woman/girl estimates focused on a single sport (soccer), but different training programme approaches (eg, neck and core strengthening vs vision training) were used. A synthesis of data evaluating neck training to reduce HAEs highlighted the heterogeneity of the current evidence base.[131] For example, interventions ranged from 5 to 27 weeks and ranged from two to four sessions per week.[131] Similarly, this review observed heterogeneity across intervention duration (6 weeks to one season) and outcomes (eg, SRC, HAE).

In community male/man/boy rugby (ages 14–18), a neuromuscular training warm-up programme performed ≥3 times per week reduced SRC rates.[132 133] Although there are no female/woman/girl-specific rugby studies, there is emerging evidence that training warm-up programmes may reduce overall injury rates in female/woman/girl soccer athletes.[134] Future training interventions assessing effect on SRC and head impacts and/or HAEs are needed.[135] Future studies should use consistent methodology and outcome measures to facilitate comparison and synthesis between studies for female/woman/girl athletes.

Policy/rule changes

Policy/rule change evaluations, which show potential benefit in some sports, continue to be primarily conducted in predominantly male/man/boy cohorts (eg, disallowing body checking in ice hockey, lowering the legal tackle height in rugby, limiting head impact exposure during American football training),[39–41 48 136 137] and thus their applicability to female/woman/girls is unknown. Based on aggregate data, there is cohort study evidence that policy disallowing body checking in paediatric ice hockey does not lead to higher rates of SRC in age groups where body checking is allowed.[39–41]

Despite that female/woman/girl-only ice hockey leagues not permitting body checking, female/woman/girl athletes are still at a higher risk of sustaining an initial and a recurrent SRC compared with male/man/boy competitions that permit body checking; moreover, girls can play with boys in leagues allowing body checking.[38 55 138] Initially, head contact rule changes did not reduce SRC or head impact rates within a male/boy cohort.[54 139] However, within a similar cohort, increasing the severity of head contact penalties was associated with lower direct head contact rates.[55] Training referees to identify and penalise head contacts may be a potential opportunity for future prevention intervention evaluation within female/woman/girl cohorts.[13]

To reduce SRC in rugby union, lowering the maximum legal tackle height was suggested to decrease the likelihood that the heads of the tackler and ball-carrier would result in head-to-head contact.[136 137] This rule change was based on professional male/man evidence.[136 137 140–143] Lowering the tackle height and reducing shared head-space between tackler and ball-carrier are equally relevant in female/woman/girl's rugby. Although no data exists within rugby union, early evidence within a professional female/women's rugby league cohort suggests a greater risk of a head injury event to both the tackler and ball-carrier when head proximity to the opposing player is above the sternum.[8] Two male/man cohorts provide limited evidence for reducing the legal tackle height to the level of the armpit.[136 137] Further research to evaluate this type of rule change in female/woman/girl athletes is warranted as differences in tackle characteristics may exist between female/woman/girl and male/man/boy rugby players.[140 144 145] Given the emerging knowledge surrounding tackle characteristics as a MRF, such a rule change could be evaluated in female/woman/girl rugby to ensure the consideration of sex/gender-specific factors.

There is a notable rise in female/woman/girl contact and/or collision sport participation, such as in rugby, Australian football and American football.[14–17] As such, exploring policies to reduce head impact/HAE exposure while still undergoing appropriate tackle training may be an opportunity to reduce SRC rates in female/woman/girl athletes. Policies limiting the number and duration of contact practices during American football led to reduced SRC and head impact/HAE rates in males/boys;[146–149] however, similar policies were not as effective in collegiate American football males/men adults.[150–152] With no evidence to date, evaluation of contact training skill development in female/woman/girl collision sports aimed to reduce SRCs and head impacts/HAEs should be evaluated.

Personal protective equipment (PPE)

Using PPE may be a relatively efficient solution for SRC prevention. Presently, headgear (hard-shell or soft-shell), eyewear and mouthguards have been evaluated within female/woman/girl cohorts. Headgear is an alternative to traditional helmets (eg, American tackle football, ice hockey) and can be used in rugby, lacrosse, Australian football and soccer. In females/women/girls lacrosse, a policy mandating headgear (hard-shell) use demonstrated lower SRC rates following its introduction,[57 64] despite concern from the lacrosse community regarding potential impacts on game performance.[153] Soft-shell headgear use in soccer across aggregated and disaggregated female/women/girl studies demonstrated inconsistent findings for protection against SRC.[61 73] Given the lack of evidence to date in female/woman/girl cohorts, hard-shell and soft-shell headgear were combined for the purposes of a meta-analysis to inform the association between headgear use and SRC. However, combining these two types of headgear should be interpreted with caution given differences in materials for these types of headgear and mechanistic implications of their protection against SRC and their use in their respective sports. Use in rugby (union, 7s) and Australian football with aggregated female/women/girl and male/man/boy groups demonstrated no protective effect on SRC[70 72 154] or SRC severity for rugby players.[70] Moreover, headgear use in male/man/boy adolescent and professional rugby union showed no significant protective effect for SRC.[13 155 156] Although the current meta-analysis displayed a potential protective effect with headgear use against SRC for female/girl athletes, the level of certainty surrounding the evidence was very low and the

heterogeneity across the studies included in the meta-analysis varied considerably. These findings are very likely to change with future high-quality research and should be interpreted with caution.

In high school female/girl's field hockey, there was no association between eyewear protection mandates and SRC rates.[67 68] Conversely, in female/girl's lacrosse, an eyewear policy may increase concussion rates.[69] Importantly, in both studies, lower rates of eye, head and face injuries were reported when eyewear protection was mandated.[67–69] Based on the current evidence, use of protective eyewear cannot be recommended for protection against SRC, but may be recommended for protection against eye and other head/face injuries.

Wearing a mouthguard is effective to prevent dental and/orofacial injuries across multiple sports, settings and populations.[157] However, for SRC, findings indicating protective effects are inconsistent.[157] Aggregate data (sex/gender) of adolescent and collegiate ice hockey players demonstrated mouthguard use was protective of SRC.[60 66 79] Without disaggregated female/woman/girl-specific data, it is uncertain if there would be similar findings in a female/woman/girl population. One study suggests higher symptom scores at 1 week post-SRC;[78] however, individuals with SRC history may be more likely to wear a mouthguard, which might explain these findings. Neck muscle strength and activation are postulated to play a role in mouthguard effectiveness against SRC (via teeth clenching).[158–160] Female/woman/girl neck strength may be weaker than male/man/boy neck strength.[6] Therefore, it is critical to explore the female/woman/girl-specific relationship between mouthguard use and SRC prior to making recommendations for mouthguard use in female/woman/girl athletes.[13]

Hard-shell helmets protect against all types of traumatic brain injuries (including SRC) across many sports,[13] albeit primarily evaluated in male/man/boy cohorts in ice hockey and American football. A structured helmet fit criteria (aggregate data) designed for youth ice hockey players showed that missing two or more (of 12) helmet fit criteria was associated with increased SRC odds,[63] but helmet age was not associated with SRC risk.[66] Helmet fit criteria should be evaluated in a female/girl-specific ice hockey context.

Management

There are studies highlighting aggregate data examining the effectiveness of policy inclusive of concussion education, protocols and medical coverage on reducing recurrent concussion rates and improving time to recovery based on aggregate data across female/woman/girl and male/man/boy athlete data.[81–87 89–91] In some studies, a reduction in recurrent concussion rates and reduced time to recovery is observed. However, the presence of medical personnel may lead to higher detection based on reporting behaviours associated with presence of medical personnel.[88]

Modifiable risk factors

This systematic review found mixed evidence for potential MRFs for SRCs and head impact/HAEs. Of the 34 risk factors identified, only field surface type was explored across three or more studies with a similar outcome for female/woman/girls,[105 108 115] and thus only one meta-analysis and semi-quantitative MRF analysis was conducted. Within this meta-analysis, there was no association observed between SRC and surface type (artificial turf vs grass). In contrast to our findings, a previous systematic review and meta-analysis which included primarily male and adult athlete populations reported a protective effect across American football, soccer and rugby demonstrating 40% lower rates of concussion on turf (IRR=0.6, 95% CI 0.47 to 0.76).[13] Currently, no mechanism for this difference could be postulated. However, a greater understanding of the implications of surface type and SRC across female/woman/girl sport is needed given the lack and heterogeneity of evidence.

Neck muscle function characteristics

The MRF most commonly examined in this review was neck muscle function. However, these studies had high levels of heterogeneity for outcome measurement and small sample sizes (<25 female/woman/girl athletes in some studies).[102 104 110 120] Differences in maximal isometric neck strength assessment (eg, technique—push or pull, testing position—seated or quadruped) made it challenging to compare data between studies. Factors related to symptoms of neck pain and neck function (eg, joint position sense, craniocervical flexion, strength, endurance, timing of activation) may affect SRC rates, as has been reported with cervical spine measures worsening following SRC.[161] Before neck muscle outcomes can be confidently considered as a MRF for SRC (in females/women/girls or males/men/boys), more evidence is required using standardised testing protocols that include muscle strength and muscle activation outcome measures.

Strengths and limitations

This systematic review provides a comprehensive evaluation of SRC prevention and MRF literature for female/woman/girl athletes. Given the paucity of female/woman/girl athlete-specific data, we were inclusive of studies reporting aggregate data (female/woman/girl and male/man/boy) to ensure we captured all available data informing appropriate recommendations for female/woman/girl athletes. Despite this effort, only 37% (40/108) of included studies contained female/woman/girl-specific estimates, with the remaining reporting aggregated data only. Within the studies that disaggregated female/woman/girl data, there was large heterogeneity of study outcome definitions across SRC and head impact/HAE, data collection methodology and exposure time. Within the studies that considered female/woman/girl estimates, a limited number of sports were evaluated, presenting an underrepresentation of sports within the female/woman/girl concussion prevention literature. Additionally, several studies contained very small sample sizes and some did not report the number of study participants (reported athlete-exposures only). Many studies were subject to potential selection bias given convenience sampling. Most studies did not evaluate potential effect measure modification, control for confounding variables (eg, previous concussion history, level of play), or consider clustering effects (eg, team, individual (when competing in several seasons)). In keeping with the findings from the International Consensus Statement on Concussion in Sport[10] there were no studies retrieved that included female/woman/girl Para sport athletes. Further attention and future research examining this understudied population is critical. Finally, the number of reviewers involved in screening and data extraction carries a small risk of error in data charting and interpretation.

Practical implications

This systematic review highlights the need for female/woman/girl-specific prevention strategies and MRFs. Evidence supports recommendations in adolescent female/woman/girl lacrosse and soccer athletes that headgear may reduce SRC incidence rates.

There are discrepancies in the literature evaluating the association between neck strength and (1) SRC or (2) head impact/HAEs in soccer and rugby. Interventions and outcome variability highlight the need for future research specifically evaluating the role of neck strength and function. Given the variability in injury definitions across prevention and MRF studies, research findings should be interpreted with caution and standardised definitions should be prioritised in future research.

Research implications

In summary, there are many gaps in the literature evaluating prevention strategies aimed to reduce concussion, head impacts, and/or HAEs in female/woman/girl athletes. Specifically, training strategy evaluations are limited in scope to muscle strength training strategies (eg, neck and core strengthening) in female/woman/girl soccer with heterogeneity of outcomes. To date, neuromuscular training warm-ups have only been evaluated in a male/man/boy rugby context. As such, there is a need for research evaluating female/woman/girl-specific training strategies (eg, neuromuscular training, tackle training) across a diversity of female/woman/girl contact and collisions sports (eg, ice hockey, rugby, lacrosse) aimed to reduce concussion. Regarding policy and rules, currently, all research is based on male/man/boy and/or aggregate data (eg, body checking policy in ice hockey, height of the legal tackle in rugby) and there is a need to consider rules of the game across multiple female/woman/girl sports. Research evaluating PPE in female/woman/girl athletes is limited to headgear as discussed in soccer and lacrosse. Female/woman/girl-specific data are needed to evaluate the effectiveness of mouthguards specifically in female/woman/girl collision sport (eg, ice hockey, rugby) where recommendations are currently based on effectiveness in male/boys ice hockey. Evaluation of helmet-fit criteria in female/woman/girl helmeted sports (eg, ice hockey, ringette) is also a significant gap.

Future research recommendations for evaluation of SRC and/or head impact/HAE prevention and SRC MRFs include prioritising disaggregated data for female/woman/girl-specific effect estimates. Priorities should also consider a focus on Para athletes, considerations across other equity-deserving athlete groups (eg, lower-middle income countries, rural communities, age groups), stronger experimental study designs, consideration of effect measure modification, adjustments for confounding and cluster(s) in analyses, and reporting of unintended consequences. In addition, future research should consider multifaceted approaches to evaluate SRC and head impact/HAE prevention strategies through adequately powered studies and the combined use of validated concussion and injury surveillance, video analysis and wearable technologies (eg, instrumented mouth guards).

CONCLUSION

There is minimal evidence for female/woman/girl athlete-specific SRC and head impact/HAE prevention strategies and MRFs and is limited to supporting headgear use in adolescent soccer/lacrosse. To make appropriate recommendations and to better inform prevention strategies for female/woman/girl athletes, evidence for female/women/girl-specific SRC prevention and MRFs is needed. The development, implementation, and evaluation of female/woman/girl-specific SRC prevention strategies are needed to reduce the SRC burden in this under-researched sport population.

Author affiliations
[1]Edinburgh Sports Medicine Research Centre, Institute for Sport, PE and Health Sciences, University of Edinburgh, Edinburgh, UK
[2]Sport Injury Prevention Research Centre, Faculty of Kinesiology, University of Calgary, Calgary, Alberta, Canada
[3]UK Collaborating Centre on Injury and Illness Prevention in Sport, Universities of Edinburgh and Bath, Edinburgh and Bath, UK
[4]Department of Occupational Science and Occupational Therapy, Temerty Faculty of Medicine, University of Toronto, Toronto, Ontario, Canada
[5]Carnegie School of Sport, Leeds Beckett University, Leeds, UK
[6]Rugby Football Union, Twickenham, UK
[7]NeuroRecovery Lab, GIGA-Consciousness, University of Liège, Liège, Belgium
[8]FIFA Medical, Fédération Internationale de Football Association, Zurich, Switzerland
[9]Faculty of Medicine and Health, The University of Sydney, Sydney, New South Wales, Australia
[10]Discipline of Physiotherapy, La Trobe University, Bundoora, Victoria, Australia
[11]UK Collaborating Centre on Injury and Illness Prevention in Sport, Universities of Edinburgh of Bath, Edinburgh and Bath, UK
[12]Centre for Health, and Injury & Illness Prevention in Sport, University of Bath, Bath, UK
[13]The Football Association, Burton-upon-Trent, UK
[14]University Hospitals Dorset NHS Foundation Trust, Poole, UK
[15]School of Psychology, Sport and Health Sciences, University of Portsmouth, Portsmouth, UK
[16]Department of Physical Medicine and Rehabilitation, Spaulding Rehabilitation Hospital/Harvard Medical School, Boston, Massachusetts, USA
[17]University of Michigan Concussion Center, University of Michigan, Ann Arbor, Michigan, USA
[18]Dept. of Psychiatry, University of Calgary, Calgary, Alberta, Canada
[19]Centre for Sport Leadership, Maties Sport, Stellenbosch University, Stellenbosch, South Africa
[20]Libraries and Cultural Resources, University of Calgary, Calgary, Alberta, Canada
[21]Division of Physiological Sciences, Faculty of Health Sciences, University of Cape Town, Cape Town, South Africa
[22]Australian Football League, Melbourne, Victoria, Australia
[23]La Trobe Sports and Exercise Medicine Research Centre, La Trobe University, Bundoora, Victoria, Australia
[24]Department of Physical Therapy, Faculty of Medicine, The University of British Columbia, Vancouver, British Columbia, Canada
[25]Arthritis Research Canada, Vancouver, British Columbia, Canada

Correction notice This article has been corrected since it published Online First. The title has been amended.

Acknowledgements We acknowledge funding from the International Olympic Committee to support methods authors, research assistants and biostatistical analyses for the FAIR Consensus systematic reviews. Additionally, we acknowledge the support of Dr. Zahra Premji (Health Research Librarian, University of Victoria) for her peer review of the injury concept used for our search strategy.

Contributors Authors contributions are reported following the Contributor Role Taxonomy (CRediT). Conceptualisation (CAE, KMC, KJS, IJS), methodology (all authors), software (CAE, IJS, J-MG, KAH), validation (CAE, IJS), formal analysis (J-MG, IJS, CAE), investigation (all authors), resources (CAE, IJS, JLW, KAH, KMC), data curation (CAE, IJS, JLW), writing—original draft (IJS), writing—review and editing (all authors), visualisation (IJS, J-MG), supervision (IJS, KJS, KMC, CAE), project administration (CAE, IJS), and funding acquisition (KMC, CAE). CAE is the guarantor of this work.

Funding Funding for the FAIR Consensus was provided by the International Olympic Committee to support methods authors, librarian and biostatistical support.

Competing interests Isla J. Shill is funded by a Canadian Institutes for Health Resarch Post-Doctoral Fellowship and holds research funding from World Rugby. Heather A. Shepherd is a co-investigator on grants from the Canadian Institutes of Health Research Strategy for Patient Oriented Research Evidence Alliance. She holds a Canadian Institutes of Health Research Postdoctoral Fellowship. She is an occupational therapist at and owner of A Head Start Therapy Services. Paul Eliason is supported by a One Child Every Child Post-doctoral Fellowship Award and is a data consultant to the National Hockey League and Canadian Football League. Ash T. Kolstad holds doctoral funding through the UCalgary Eyes High Doctoral Recruitment Scholarship and Izaak Walton Killam Pre-Doctoral AwardOmar Heyward is a consultant to the Rugby Football Union. His research fellowship is part-funded by Premiership Rugby.Géraldine Martens holds a Postdoctoral Researcher fellowship from the Belgian Fund for Scientific Research (F.R.S.-FNRS). She is an active member of the French-speaking Olympic Sports Medicine Research Network (ReFORM) funded by the International Olympic Committee (IOC) through the IOC Research Centres program. Kerry Peek is employed by FIFA as a medical researcher.Clara Soligon has no conflicts of interest to declare. Matthew G King is an Associate Editor for British Journal of Sports Medicine.Stephen W. West holds research funding from World Rugby.Osman Hassan Ahmed is a Senior Physiotherapist at University Hospitals Dorset NHS Foundation Trust (England) and is Para Football Physiotherapy Lead, Para Football Classification Lead, and the England Cerebral Palsy Football

squad Physiotherapist at the Football Association (England). He also works on a consultancy basis with the Football Association teaching on their Advanced Trauma and Medical Management in Football course. He has a Visiting Senior Lecturer position at the University of Portsmouth, England (unpaid). He sits on several disability sport committees, including Para Football Foundation as Medical Unit Co-Lead (unpaid), the International Federation of Cerebral Palsy Football as Medical and Sports Science Director (unpaid) and the International Blind Sports Association as Medical Committee member (unpaid). He has Associate Editor positions at the British Journal of Sports Medicine (unpaid) and BMJ Open Sports & Exercise Medicine (unpaid). He is also an Institutional Ethics Committee external member for World Rugby (unpaid) and co-chair of the Concussion in Para Sport Group (unpaid). Cheri Blauwet is an Associate Editor for BJSM and on the IPHP Editorial Board. Steve Broglio has current research funding from the National Institutes of Health; Centers for Disease Control and Prevention; Department of Defense - USA Medical Research Acquisition Activity, and National Collegiate Athletic Association. He has past research funding from the National Athletic Trainers' Association Foundation; National Football League/Under Armour/GE; Simbex; and ElmindA. He has consulted for US Soccer (paid), US Cycling (unpaid), University of Calgary SHRed Concussions external advisory board (unpaid), and is the President of the Concussion in Sport Group (unpaid). He has received compensation for medico-legal litigation, and received speaker honorarium and travel reimbursements for talks given. He is co-author of "Biomechanics of Injury (3rd edition, Human Kinetics)," holds a patents on "Brain Metabolism Monitoring Through CCO Measurements Using All-Fiber-Integrated Super-Continuum Source" (U.S. US 11, 529, 091 B2) and "Organ Or Tissue Hemodynamic And Metabolic Function Measured Using Super-Continuum Light Sources" (US 12,089,945 B2), and is on the scientific advisory board for C3Logix (stock options). He is is/was on the editorial boards (all unpaid) for Journal of Athletic Training (2015 to present), Concussion (2014 to present), Athletic Training & Sports Health Care (2008 to present), British Journal of Sports Medicine (2008 to 2019).Araba Chintoh is a member of World Rugby's Concussion Working Group (paid).Jean-Michel Galarneau has no conflicts of interest to declare.Alix Hayden has not conflicts of interest to declare.Sharief Hendricks is research consultant for World Rugby, the South African Rugby Union, and the South African Crickters's Association. He is also a BJSM Associate Editor. Michael Makdissi is a Sport and Exercise Medicine Physician working in private consulting practice. He is a consultant to the Australian Football League (AFL, Chief Medical Officer), Tennis Australia (Australian Open, Player Medical Services), World Rugby (Independent Concussion Consultant) and the English Premier League. He is a former team physician at the Hawthorn Football Club (AFL) and former Chief Executive Officer of the AFL Doctors Association. He has previously received research grants from the AFL, outside the submitted work and is a chief investigator (unpaid) on projects receiving grants from the Australian Government Department of Health Medical Research and Future Fund. He is on the editorial board of the BJSM (unpaid). He has received travel support from the AFL, FIFA and the International Olympic Committee to attend and present at international conferences. He is a member of the International Concussion in Sport Group (unpaid) and the Australian Rugby Union Concussion Advisory Group (unpaid).Debbie Palmer is principal investigator holding research funding from World Rugby, the International Olympic Committee and the Scottish Rugby Union. Stacy Sick holds doctoral funding through the Canadian Institute of Health Research Jackie L. Whittaker is an Associate Editor for BJSM and is supported by a Michael Smith Health Research British Columbia Scholar Award (SCH-2020-0403). She holds peer-reviewed funding from the Canadian Institutes of Health Research (CIHR) and Arthritis Society.Kay Crossley is project leader of the GLA:D, Australia, and holds a research grant from Levin Health outside the submitted work, and holds peer-reviewed research funding from the Australian Football League, and the Australian National Health and Medical Research Council, and Medical Research Future Fund. Kathryn J. Schneider holds research funding from World Rugby and Mitacs. She was the co-lead author of the Amsterdam International Consensus Statement on Concussion in Sport and co-chair of the Scientific Committee and Expert Panel for the 6th International Conference on Concussion in Sport. She received an educational grant paid to her institution for the administrative aspects related to the preparatory activities for the Amsterdam Consensus process. She is a physiotherapist consultant at Evidence Sport and Spinal Therapy and at the University of Calgary Sport Medicine Centre. Carolyn A. Emery is a Deputy Editor of British Journal of Sports Medicine, was a member of the Expert Panel for the 6th International Conference on Concussion in Sport and holds a Canada Research Chair (Tier 1) in Concussion and research funding from Canadian Institues of Health Research, International Olympic Committee, World Rugby, National Football League Play Safe Program, and Canada Foundation for Innovation.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. Deidentified data can be made available upon reasonable request to the corresponding author. Please email the corresponding author with a written proposal outlining the specific research aims and analysis plan and why these specific data are needed. A formal data-sharing agreement between institutions will be required.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

ORCID iDs
Isla J Shill https://orcid.org/0000-0003-0175-2354
Heather A Shepherd https://orcid.org/0000-0003-0665-8674
Paul H Eliason https://orcid.org/0000-0001-6985-7043
Ash T Kolstad https://orcid.org/0000-0002-2994-9880
Omar Heyward https://orcid.org/0000-0002-7390-6511
Géraldine Martens https://orcid.org/0000-0001-7038-7165
Kerry Peek https://orcid.org/0000-0002-2194-3353
Clara A Soligon https://orcid.org/0009-0002-1795-1120
Matthew G King https://orcid.org/0000-0003-0470-5924
Osman Hassan Ahmed https://orcid.org/0000-0002-1439-0076
Steven Broglio https://orcid.org/0000-0002-2282-9325
Araba Chintoh https://orcid.org/0000-0002-7775-5810
Jean-Michel Galarneau https://orcid.org/0000-0003-3784-8753
K Alix Hayden https://orcid.org/0000-0002-0057-1327
Sharief Hendricks https://orcid.org/0000-0002-3416-6266
Michael Makdissi https://orcid.org/0000-0003-0334-7133
Debbie Palmer https://orcid.org/0000-0002-4676-217X
Jackie L Whittaker https://orcid.org/0000-0002-6591-4976
Kay M Crossley https://orcid.org/0000-0001-5892-129X
Kathryn J Schneider https://orcid.org/0000-0002-5951-5899
Carolyn A Emery https://orcid.org/0000-0002-9499-6691

REFERENCES

1 McGroarty NK, Brown SM, Mulcahey MK. Sport-Related Concussion in Female Athletes: A Systematic Review. *Orthop J Sports Med* 2020;8:2325967120932306.
2 Covassin T, Moran R, Elbin RJ. Sex Differences in Reported Concussion Injury Rates and Time Loss From Participation: An Update of the National Collegiate Athletic Association Injury Surveillance Program From 2004-2005 Through 2008-2009. *J Athl Train* 2016;51:189–94.
3 Shill IJ, West SW, Sick S, *et al*. Differences in injury and concussion rates in a cohort of Canadian female and male youth Rugby Union: a step towards targeted prevention strategies. *Br J Sports Med* 2024;58:34–41.
4 Resch JE, Rach A, Walton S, *et al*. Sport Concussion and the Female Athlete. *Clin Sports Med* 2017;36:717–39.
5 Covassin T, Elbin RJ. The female athlete: the role of gender in the assessment and management of sport-related concussion. *Clin Sports Med* 2011;30:125–31, .
6 Williams EMP, Petrie FJ, Pennington TN, *et al*. Sex differences in neck strength and head impact kinematics in university rugby union players. *Eur J Sport Sci* 2022;22:1649–58.
7 Caccese JB, Lamond LC, Buckley TA, *et al*. Reducing purposeful headers from goal kicks and punts may reduce cumulative exposure to head acceleration. *Res Sports Med* 2016;24:407–15.
8 McLeod S, Tucker R, Edwards S, *et al*. A case-control study of tackle based head impact event (HIE) risk factors from the first three seasons of the National Rugby League Women's competition. *Front Sports Act Living* 2023;5:1080356.
9 Emery CA, Pasanen K. Current trends in sport injury prevention. *Best Pract Res Clin Rheumatol* 2019;33:3–15.
10 Patricios JS, Schneider KJ, Dvorak J, *et al*. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport-Amsterdam, October 2022. Br J Sports Med 2022;57:695–711.
11 Schneider KJ, Emery CA, Black A, *et al*. Adapting the Dynamic, Recursive Model of Sport Injury to Concussion: An Individualized Approach to Concussion Prevention, Detection, Assessment, and Treatment. *J Orthop Sports Phys Ther* 2019;49:799–810.
12 Meeuwisse WH, Tyreman H, Hagel B, *et al*. A dynamic model of etiology in sport injury: the recursive nature of risk and causation. *Clin J Sport Med* 2007;17:215–9.

13 Eliason PH, Galarneau J-M, Kolstad AT, *et al*. Prevention strategies and modifiable risk factors for sport-related concussions and head impacts: a systematic review and meta-analysis. *Br J Sports Med* 2023;57:749–61.
14 National Football League Football Operations. More girls are playing tackle football. 2020. Available: https://operations.nfl.com/gameday/analytics/stats-articles/more-girls-are-playing-tackle-football/#:~:text=Playing%20Tackle%20Football-,January%209%2C%202020,girls%20participation%20continues%20to%20grow
15 Women's football alliance. 2025 Available: https://wfaprofootball.com
16 World Rugby. Year in review. Dublin, Ireland, 2021.
17 Australian Football League. 2023 annual report. 2023.
18 Higgins JPT, Thomas J, Chandler J, *et al*. *Cochrane handbook for systematic reviews of interventions*. 2nd edn. John Wiley & Sons, 2019.
19 Page MJ, McKenzie JE, Bossuyt PM, *et al*. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *J Clin Epidemiol* 2021;134:178–89.
20 Rethlefsen ML, Kirtley S, Waffenschmidt S, *et al*. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Syst Rev* 2021;10:39.
21 Veritas Health Innovation. Covidence systematic review software. Melbourne, Australia,
22 Downs SH, Black N. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. *J Epidemiol Community Health* 1998;52:377–84.
23 StataCorp. Stata statistical software: release 18. college station, tx: statacorp llc. 2023.
24 Huguet A, Hayden JA, Stinson J, *et al*. Judging the quality of evidence in reviews of prognostic factor research: adapting the GRADE framework. *Syst Rev* 2013;2:71.
25 Antonoff DG, Goss J, Langevin TL, *et al*. Unexpected Findings from a Pilot Study on Vision Training as a Potential Intervention to Reduce Subconcussive Head Impacts during a Collegiate Ice Hockey Season. *J Neurotrauma* 2021;38:1783–90.
26 Howell DR, Seehusen CN, Carry PM, *et al*. An 8-Week Neuromuscular Training Program After Concussion Reduces 1-Year Subsequent Injury Risk: A Randomized Clinical Trial. *Am J Sports Med* 2022;50:1120–9.
27 Mansell J, Tierney RT, Sitler MR, *et al*. Resistance training and head-neck segment dynamic stabilization in male and female collegiate soccer players. *J Athl Train* 2005;40:310–9.
28 Morrissey S, Dumire R, Causer T, *et al*. The missing piece of the concussion discussion: primary prevention of mild traumatic brain injury in student athletes. *J Emerg Crit Care Med* 2019;3:8.
29 Müller C, Zentgraf K. Neck and Trunk Strength Training to Mitigate Head Acceleration in Youth Soccer Players. *J Strength Cond Res* 2021;35:S81–9.
30 Peek K, Andersen J, McKay MJ, *et al*. The Effect of the FIFA 11 + with Added Neck Exercises on Maximal Isometric Neck Strength and Peak Head Impact Magnitude During Heading: A Pilot Study. *Sports Med* 2022;52:655–68.
31 Peek K, Versteegh T, Veith S, *et al*. Injury-Reduction Programs Containing Neuromuscular Neck Exercises and the Incidence of Soccer-Related Head and Neck Injuries. *J Athl Train* 2023;58:519–27.
32 Wahlquist VE, Glutting JJ, Kaminski TW. Examining the influence of the Get aHEAD Safely in Soccer TM program on head impact kinematics and neck strength in female youth soccer players. *Res Sports Med* 2024;32:17–27.
33 Wahlquist VE, Kaminski TW. Analysis of Head Impact Biomechanics in Youth Female Soccer Players Following the Get aHEAD Safely in Soccer TM Heading Intervention. *Sensors (Basel)* 2021;21:3859.
34 Waring KM, Smith ER, Austin GP, *et al*. Exploring the Effects of a Neck Strengthening Program on Purposeful Soccer Heading Biomechanics and Neurocognition. *Int J Sports Phys Ther* 2022;17:1043–52.
35 Black AM, Hagel BE, Palacios-Derflingher L, *et al*. The risk of injury associated with body checking among Pee Wee ice hockey players: an evaluation of Hockey Canada's national body checking policy change. *Br J Sports Med* 2017;51:1767–72.
36 Black AM, Macpherson AK, Hagel BE, *et al*. Policy change eliminating body checking in non-elite ice hockey leads to a threefold reduction in injury and concussion risk in 11- and 12-year-old players. *Br J Sports Med* 2016;50:55–61.
37 Eliason P, Galarneau J-M, Shill I, *et al*. Factors Associated With Concussion Rates in Youth Ice Hockey Players: Data From the Largest Longitudinal Cohort Study in Canadian Youth Ice Hockey. *Clin J Sport Med* 2023;33:497–504.
38 Eliason P, Galarneau J-M, Schneider K, *et al*. Higher hazard of concussion combined with recurrent concussion in female youth ice hockey compared with males: An opportunity for concussion prevention. *Sports Orthopaedics and Traumatology* 2023;39:113–23.
39 Eliason P, Hagel BE, Palacios-Derflingher L, *et al*. No association found between body checking experience and injury or concussion rates in adolescent ice hockey players. *Br J Sports Med* 2022;56:1337–44.
40 Eliason PH, Hagel BE, Palacios-Derflingher L, *et al*. Bodychecking experience and rates of injury among ice hockey players aged 15-17 years. *CMAJ* 2022;194:E834–42.
41 Emery C, Kang J, Shrier I, *et al*. Risk of injury associated with bodychecking experience among youth hockey players. *Can Med Assoc J* 2011;183:1249–56.
42 Emery C, Palacios-Derflingher L, Black AM, *et al*. Does disallowing body checking in non-elite 13- to 14-year-old ice hockey leagues reduce rates of injury and concussion? A cohort study in two Canadian provinces. *Br J Sports Med* 2020;54:414–20.
43 Emery CA, Eliason P, Warriyar V, *et al*. Body checking in non-elite adolescent ice hockey leagues: it is never too late for policy change aiming to protect the health of adolescents. *Br J Sports Med* 2022;56:12–7.
44 Emery CA, Kang J, Schneider KJ, *et al*. Risk of injury and concussion associated with team performance and penalty minutes in competitive youth ice hockey. *Br J Sports Med* 2011;45:1289–93.
45 Emery CA, Kang J, Shrier I, *et al*. Risk of injury associated with body checking among youth ice hockey players. *JAMA* 2010;303:2265–72.
46 Hagel BE, Marko J, Dryden D, *et al*. Effect of bodychecking on injury rates among minor ice hockey players. *CMAJ* 2006;175:155–60.
47 Jae-Ok K, Voaklander D. Effects of Competition Rule Changes on the Incidence of Head Kicks and Possible Concussions in Taekwondo. *Clin J Sport Med* 2016;26:239–44.
48 Kolstad AT, Nadeau L, Eliason PH, *et al*. Does disallowing body checking impact offensive performance in non-elite under-15 and under-18 youth ice hockey leagues? A video-analysis study. *Int J Sports Sci Coach* 2022.
49 Lalji R, Snider H, Chow N, *et al*. The 2015 U.S. Soccer Federation header ban and its effect on emergency room concussion rates in soccer players aged 10-13. *J Can Chiropr Assoc* 2020;64:187–92.
50 Macan J, Bundalo-Vrbanac D, Romić G. Effects of the new karate rules on the incidence and distribution of injuries. *Br J Sports Med* 2006;40:326–30; .
51 Nadkarni L, Haskins A, Holt C, *et al*. Reduction of High School Ice Hockey Injuries With Implementation of New Checking/Boarding Rules. *Clin J Sport Med* 2021;31:e420–4.
52 Obana KK, Mueller JD, Saltzman BM, *et al*. Targeting Rule Implementation Decreases Concussions in High School Football: A National Concussion Surveillance Study. *Orthop J Sports Med* 2021;9:23259671211031191.
53 Smith AM, Gaz DV, Larson D, *et al*. Does fair play reduce concussions? A prospective, comparative analysis of competitive youth hockey tournaments. *BMJ Open Sport Exerc Med* 2016;2:e000074.
54 Williamson RA, Kolstad AT, Krolikowski M, *et al*. Incidence of Head Contacts, Penalties, and Player Contact Behaviors in Youth Ice Hockey: Evaluating the "Zero Tolerance for Head Contact" Policy Change. *Orthop J Sports Med* 2021;9:2325967121992375.
55 Williamson RA, Kolstad AT, Nadeau L, *et al*. Does Increasing the Severity of Penalties Assessed in Association With the "Zero Tolerance for Head Contact" Policy Translate to a Reduction in Head Impact Rates in Youth Ice Hockey?. Clin J Sport Med 2022;32:e598–604.
56 Barbic D, Pater J, Brison RJ. Comparison of mouth guard designs and concussion prevention in contact sports: a multicenter randomized controlled trial. *Clin J Sport Med* 2005;15:294–8.
57 Baron SL, Veasley SJ, Kingery MT, *et al*. Decreased Injury Rate Following Mandated Headgear Use in Women's Lacrosse. *Bull Hosp Jt Dis (2013)* 2013;78:260–5.
58 Castellanos J, Phoo CP, Eckner JT, *et al*. Predicting Risk of Sport-Related Concussion in Collegiate Athletes and Military Cadets: A Machine Learning Approach Using Baseline Data from the CARE Consortium Study. *Sports Med* 2021;51:567–79.
59 Caswell SV, Kelshaw PM, Lincoln AE, *et al*. The Effects of Headgear in High School Girls' Lacrosse. *Orthop J Sports Med* 2020;8.
60 Chisholm DA, Black AM, Palacios-Derflingher L, *et al*. Mouthguard use in youth ice hockey and the risk of concussion: nested case-control study of 315 cases. *Br J Sports Med* 2020;54:866–70.
61 Delaney JS, Al-Kashmiri A, Drummond R, *et al*. The effect of protective headgear on head injuries and concussions in adolescent football (soccer) players. *Br J Sports Med* 2008;42:110–5.
62 Filbay SR, Bullock GS, Sanchez-Santos MT, *et al*. A Higher Playing Standard, Bowling, and Intermittent Helmet Use Are Related to a Greater Odds of Injury or Concussion in Cricket. *Clin J Sport Med* 2022;32:e300–7.
63 Gamble ASD, Bigg JL, Sick S, *et al*. Helmet Fit Assessment and Concussion Risk in Youth Ice Hockey Players: A Nested Case-Control Study. *J Athl Train* 2021;56:845–50.
64 Herman DC, Caswell SV, Kelshaw PM, *et al*. Association of headgear mandate and concussion injury rates in girls' high school lacrosse. *Br J Sports Med* 2022;56:970–4.
65 Hurst HT, Rylands L, Atkins S, *et al*. Profiling of translational and rotational head accelerations in youth BMX with and without neck brace. *J Sci Med Sport* 2018;21:263–7.
66 Kolstad AT, Eliason PH, Galarneau J-M, *et al*. Protective equipment in youth ice hockey: are mouthguards and helmet age relevant to concussion risk? *Br J Sports Med* 2023;57:571–7.
67 Kriz PK, Comstock RD, Zurakowski D, *et al*. Effectiveness of protective eyewear in reducing eye injuries among high school field hockey players. *Pediatrics* 2012;130:1069–75.
68 Kriz PK, Zurakowski D, Almquist JL, *et al*. Eye Protection and Risk of Eye Injuries in High School Field Hockey. *Pediatrics* 2015;136:521–7.

69 Lincoln AE, Caswell SV, Almquist JL, *et al*. Effectiveness of the Women's Lacrosse Protective Eyewear Mandate in the Reduction of Eye Injuries. *Am J Sports Med* 2012;40:611–4.
70 Lopez Jr V, Ma R, WeinsteinMG, *et al*. Concussive Injuries in Rugby 7s: An American Experience and Current Review. *Medicine & Science in Sports & Exercise* 2016;48:1320–30.
71 Makovec Knight J, Mitra B, McIntosh A, *et al*. The association of padded headgear with concussion and injury risk in junior Australian football: A prospective cohort study. *J Sci Med Sport* 2022;25:312–20.
72 Marshall SW, Loomis DP, Waller AE, *et al*. Evaluation of protective equipment for prevention of injuries in rugby union. *Int J Epidemiol* 2005;34:113–8.
73 McGuine T, Post E, Pfaller AY, *et al*. Does soccer headgear reduce the incidence of sport-related concussion? A cluster, randomised controlled trial of adolescent athletes. *Br J Sports Med* 2020;54:408–13.
74 Mihalik JP, McCaffrey MA, Rivera EM, *et al*. Effectiveness of mouthguards in reducing neurocognitive deficits following sports-related cerebral concussion. *Dent Traumatol* 2007;23:14–20.
75 Ono Y, Tanaka Y, Sako K, *et al*. Association between Sports-Related Concussion and Mouthguard Use among College Sports Players: A Case-Control Study Based on Propensity Score Matching. *IJERPH* 2020;17:4493.
76 Raaii F, Vaidya N, Vaidya K, *et al*. Patterns of Mouthguard Utilization Among Atom and Pee Wee Minor Ice Hockey Players: A Pilot Study. *Clin J Sport Med* 2011;21:320–4.
77 Tierney RT, Higgins M, Caswell SV, *et al*. Sex differences in head acceleration during heading while wearing soccer headgear. *J Athl Train* 2008;43:578–84.
78 van Ierssel J, Ledoux A-A, Tang K, *et al*. Sex-Based Differences in Symptoms With Mouthguard Use After Pediatric Sport-Related Concussion. *J Athl Train* 2021;56:1188–96.
79 Van Pelt KL, Caccese JB, Eckner JT, *et al*. Detailed description of Division I ice hockey concussions: Findings from the NCAA and Department of Defense CARE Consortium. *J Sport Health Sci* 2021;10:162–71.
80 Willmott C, Reyes J, K Nguyen JV, *et al*. A pilot case crossover study of the use of padded headgear in junior Australian football. *Concussion* 2022;7:CNC99.
81 Arakkal AT, Barón AE, Lamb MM, *et al*. Evaluating the effectiveness of traumatic brain injury state laws among high school athletes. *Inj Epidemiol* 2020;7:12.
82 Asken BM, Bauer RM, Guskiewicz KM, *et al*. Immediate Removal From Activity After Sport-Related Concussion Is Associated With Shorter Clinical Recovery and Less Severe Symptoms in Collegiate Student-Athletes. *Am J Sports Med* 2018;46:1465–74.
83 Asken BM, McCrea MA, Clugston JR, *et al*. "Playing Through It": Delayed Reporting and Removal From Athletic Activity After Concussion Predicts Prolonged Recovery. *J Athl Train* 2016;51:329–35.
84 Baugh CM, Meehan WP, McGuire TG, *et al*. Staffing, Financial, and Administrative Oversight Models and Rates of Injury in Collegiate Athletes. *J Athl Train* 2020;55:580–6.
85 Charek DB, Elbin RJ, Sufrinko A, *et al*. Preliminary Evidence of a Dose-Response for Continuing to Play on Recovery Time After Concussion. *J Head Trauma Rehabil* 2020;35:85–91.
86 Emery CA, Warriyar Kv V, Black AM, *et al*. Factors Associated With Clinical Recovery After Concussion in Youth Ice Hockey Players. *Orthop J Sports Med* 2021;9:23259671211013370.
87 Miller GF, Sarmiento K, Haarbauer-Krupa J, *et al*. The Association Between School District-Based Policies Related to Concussions and Concussions Among High School Students. *J Sch Health* 2022;92:140–7.
88 Pierpoint LA, LaBella CR, Collins CL, *et al*. Injuries in girls' soccer and basketball: a comparison of high schools with and without athletic trainers. *Inj Epidemiol* 2018;5:29.
89 Yang J, Comstock RD, Yi H, *et al*. New and Recurrent Concussions in High-School Athletes Before and After Traumatic Brain Injury Laws, 2005-2016. *Am J Public Health* 2017;107:1916–22.
90 Yang J, Harvey HH, Sullivan L, *et al*. Association Between Design Elements of Concussion Laws and Reporting of Sports-Related Concussions Among US High School Athletes, 2009-2017. *Public Health Rep* 2021;136:745–53.
91 Zynda AJ, Worrall HM, Sabatino MJ, *et al*. Continued play following adolescent sport-related concussion: Prospective data from the North Texas Concussion Registry (ConTex). *Appl Neuropsychol Child* 2022;11:740–51.
92 Ali M, Dreher N, Hannah T, *et al*. Concussion Incidence and Recovery Among Youth Athletes With ADHD Taking Stimulant-Based Therapy. *Orthop J Sports Med* 2021;9:23259671211032564.
93 Arriaza R, Cierna D, Regueiro P, *et al*. Low risk of concussions in top-level karate competition. *Br J Sports Med* 2017;51:226–30.
94 Augustovičová D, Lystad RP, Arriaza R. Time-Loss Injuries in Karate: A Prospective Cohort Study of 4 Consecutive World Karate Championships. *Orthop J Sports Med* 2019;7:2325967119865866.
95 Baker M, Quesnele J, Baldisera T, *et al*. Exploring the role of cervical spine endurance as a predictor of concussion risk and recovery following sports related concussion. *Musculoskeletal Science and Practice* 2019;42:193–7.
96 Biese KM, Winans M, Fenton AN, *et al*. High School Sport Specialization and Injury in Collegiate Club-Sport Athletes. *J Athl Train* 2021;56:1271–7.
97 Caccese JB, Buckley TA, Tierney RT, *et al*. Head and neck size and neck strength predict linear and rotational acceleration during purposeful soccer heading. *Sports Biomech* 2018;17:462–76.
98 Collins CL, Fletcher EN, Fields SK, *et al*. Neck strength: a protective factor reducing risk for concussion in high school sports. *J Prim Prev* 2014;35:309–19.
99 Curran-Sills G, Abedin T. Risk factors associated with injury and concussion in sanctioned amateur and professional mixed martial arts bouts in Calgary, Alberta. *BMJ Open Sport Exerc Med* 2018;4:e000348.
100 Denham BE. Associations Between Sports and Exercise Participation, Use of Narcotic Pain Relievers, and Concussions in High-School Seniors. *J Sport Behav* 2022;45:17–30.
101 DePadilla L, Miller GF, Jones SE, *et al*. Substance Use and Sports- or Physical Activity-Related Concussions Among High School Students. *J Sch Nurs* 2022;38:511–8.
102 Dezman ZDW, Ledet EH, Kerr HA. Neck strength imbalance correlates with increased head acceleration in soccer heading. *Sports Health* 2013;5:320–6.
103 Eagle SR, Asken B, Trbovich A, *et al*. Estimated Duration of Continued Sport Participation Following Concussions and Its Association with Recovery Outcomes in Collegiate Athletes: Findings from the NCAA/DoD CARE Consortium. *Sports Med* 2022;52:1991–2001.
104 Eckner JT, Oh YK, Joshi MS, *et al*. Effect of neck muscle strength and anticipatory cervical muscle activation on the kinematic response of the head to impulsive loads. *Am J Sports Med* 2014;42:566–76.
105 Ekstrand J, Hägglund M, Fuller CW. Comparison of injuries sustained on artificial turf and grass by male and female elite football players. *Scand J Med Sci Sports* 2011;21:824–32.
106 Esopenko C, de Souza N, Conway F, *et al*. Bigger Necks Are Not Enough: An Examination of Neck Circumference in Incoming College Athletes. *J Prim Prev* 2020;41:421–9.
107 Filben TM, Tomblin BT, Pritchard NS, *et al*. Assessing the association between on-field heading technique and head impact kinematics in a cohort of female youth soccer players. *Sci Med Footb* 2024;8:341–50.
108 Fuller CW, Dick RW, Corlette J, *et al*. Comparison of the incidence, nature and cause of injuries sustained on grass and new generation artificial turf by male and female football players. Part 2: training injuries. *Br J Sports Med* 2007;41 Suppl 1:i27–32.
109 Gallagher V, Kramer N, Abbott K, *et al*. The Effects of Sex Differences and Hormonal Contraception on Outcomes after Collegiate Sports-Related Concussion. *J Neurotrauma* 2018;35:1242–7.
110 Gutierrez GM, Conte C, Lightbourne K. The relationship between impact force, neck strength, and neurocognitive performance in soccer heading in adolescent females. *Pediatr Exerc Sci* 2014;26:33–40.
111 Harriss A, Johnson AM, Walton DM, *et al*. Head impact magnitudes that occur from purposeful soccer heading depend on the game scenario and head impact location. *Musculoskelet Sci Pract* 2019;40:53–7.
112 Koh JO, Cassidy JD. Incidence Study of Head Blows and Concussions in Competition Taekwondo. *Clin J Sport Med* 2004;14:72–9.
113 Kriz PK, Stein C, Kent J, *et al*. Physical Maturity and Concussion Symptom Duration among Adolescent Ice Hockey Players. *J Pediatr* 2016;171:234–9.
114 Li AY, Durbin JR, Hannah TC, *et al*. High altitude modulates concussion incidence, severity, and recovery in young athletes. *Brain Inj* 2022;36:733–9.
115 Meyers MC. Incidence, mechanisms, and severity of match-related collegiate women's soccer injuries on FieldTurf and natural grass surfaces: a 5-year prospective study. *Am J Sports Med* 2013;41:2409–20.
116 Okoruwa ET, Abbott A, Stamm M, *et al*. Sport Specialization Classification and Injury Risk in Female Athletes. *Sports Health* 2022;14:797–804.
117 Peek K, McKay M, Fu A, *et al*. The effect of ball characteristics on head acceleration during purposeful heading in male and female youth football players. *Sci Med Footb* 2021;5:195–203.
118 Raikes AC, Athey A, Alfonso-Miller P, *et al*. Insomnia and daytime sleepiness: risk factors for sports-related concussion. *Sleep Med* 2019;58:66–74.
119 Reed N, Taha T, Monette G, *et al*. A Preliminary Exploration of Concussion and Strength Performance in Youth Ice Hockey Players. *Int J Sports Med* 2016;37:708–13.
120 Reha T, McNabb C, Netto K, *et al*. Head Accelerations during a 1-on-1 Rugby Tackling Drill Performed by Experienced Rugby Union Players. *Brain Sci* 2021;11:1497.
121 Riegler KE, Guty ET, Thomas GA, *et al*. Prospective Implications of Insufficient Sleep for Athletes. *J Athl Train* 2023;58:414–22.
122 Rogerson CE, Jackson BC, Breedlove KM, *et al*. The Effects of Preparedness and Activity on Head Impacts in Lacrosse Athletes. *International Journal of Athletic Therapy and Training* 2023;28:239–43.
123 Sarmiento K, Miller GF, Jones SE. Sports-Related Concussions and Adverse Health Behaviors Among Middle and High School Students. *Am J Sports Med* 2023;51:503–10.
124 Schulz MR, Marshall SW, Mueller FO, *et al*. Incidence and risk factors for concussion in high school athletes, North Carolina, 1996-1999. *Am J Epidemiol* 2004;160:937–44.

125 Sherman K, Tyree P, Ford AI, *et al*. Association between steroid use and concussions among high school athletes: a cross-sectional analysis of the Youth Risk Behavior Surveillance System. *J Osteopath Med* 2024;124:27–34.
126 Smith DW, Myer GD, Currie DW, *et al*. Altitude Modulates Concussion Incidence: Implications for Optimizing Brain Compliance to Prevent Brain Injury in Athletes. *Orthop J Sports Med* 2013;1:2325967113511588.
127 Spiera Z, Hannah T, Li A, *et al*. Nonsteroidal anti-inflammatory drug use and concussions in adolescent athletes: incidence, severity, and recovery. *J Neurosurg Pediatr* 2021;28:476–82.
128 Stockbridge MD, Keser Z, Newman RS. Concussion in Women's Flat-Track Roller Derby. *Front Neurol* 2022;13:809939.
129 Whatman C, van den Berg C, Black AM, *et al*. High Sport Specialization Is Associated With More Musculoskeletal Injuries in Canadian High School Students. *Clin J Sport Med* 2023;33:233–8.
130 National Institute of Health. NIH study qualtiy assessment tools. 2021. Available: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools
131 Fownes-Walpole M, Heyward O, Till K, *et al*. Combining evidence and practice to optimise neck training aimed at reducing head acceleration events in sport: a systematic review and Delphi-consensus study. *Br J Sports Med* 2025;59:1027–42.
132 Hislop MD, Stokes KA, Williams S, *et al*. Reducing musculoskeletal injury and concussion risk in schoolboy rugby players with a pre-activity movement control exercise programme: a cluster randomised controlled trial. *Br J Sports Med* 2017;51:1140–6.
133 Attwood MJ, Roberts SP, Trewartha G, *et al*. Efficacy of a movement control injury prevention programme in adult men's community rugby union: a cluster randomised controlled trial. *Br J Sports Med* 2018;52:368–74.
134 Soligard T, Myklebust G, Steffen K, *et al*. Comprehensive warm-up programme to prevent injuries in young female footballers: Cluster randomised controlled trial. *BMJ* 2009;338:95–9.
135 Patterson BE, Donaldson A, Cowan SM, *et al*. Evaluation of an injury prevention programme (Prep-to-Play) in women and girls playing Australian Football: design of a pragmatic, type III, hybrid implementation-effectiveness, stepped-wedge, cluster randomised controlled trial. *BMJ Open* 2022;12:e062483.
136 Stokes KA, Locke D, Roberts S, *et al*. Does reducing the height of the tackle through law change in elite men's rugby union (The Championship, England) reduce the incidence of concussion? A controlled study in 126 games. *Br J Sports Med* 2021;55:220–5.
137 van Tonder R, Starling L, Surmon S, *et al*. Tackling sport-related concussion: effectiveness of lowering the maximum legal height of the tackle in amateur male rugby - a cross-sectional analytical study. *Inj Prev* 2023;29:56–61.
138 Heming EE, Sobry AJ, Cairo AL, *et al*. Higher Rates of Head Contacts, Body Checking, and Suspected Injuries in Ringette Than Female Ice Hockey: Time to Ring in Opportunities for Prevention. *Clin J Sport Med* 2023;33:151–6.
139 Krolikowski MP, Black AM, Palacios-Derflingher L, *et al*. The Effect of the "Zero Tolerance for Head Contact" Rule Change on the Risk of Concussions in Youth Ice Hockey Players. *Am J Sports Med* 2017;45:468–73.
140 Cross MJ, Tucker R, Raftery M, *et al*. Tackling concussion in professional rugby union: a case–control study of tackle-based risk factors and recommendations for primary prevention. *Br J Sports Med* 2019;53:1021–5.
141 Tucker R, Raftery M, Fuller GW, *et al*. A video analysis of head injuries satisfying the criteria for a head injury assessment in professional Rugby Union: a prospective cohort study. *Br J Sports Med* 2017;51:1147–51.
142 Tucker R, Raftery M, Kemp S, *et al*. Risk factors for head injury events in professional rugby union: a video analysis of 464 head injury events to inform proposed injury prevention strategies. *Br J Sports Med* 2017;51:1152–7.
143 Rugby Football Union. Rationale and evidence for proposal to reduce the tackle height in england's community game. London, UK, 2023.
144 Shill IJ, Galarneau J-M, Hendricks S, *et al*. Tackle characteristics associated with suspected concussion in female varsity rugby union: a case-control video analysis study. *Int J Perform Anal Sport* 2025;25:386–404.
145 Dane K, West SW, Hendricks S, *et al*. Are we tackle ready? Cross-sectional video analysis of match tackle characteristics in elite women's Rugby Union. *Eur J Sport Sci* 2024;24:999–1009.
146 Broglio SP, Martini D, Kasper L, *et al*. Estimation of head impact exposure in high school football: implications for regulating contact practices. *Am J Sports Med* 2013;41:2877–84.
147 Broglio SP, Williams RM, O'Connor KL, *et al*. Football Players' Head-Impact Exposure After Limiting of Full-Contact Practices. *J Athl Train* 2016;51:511–8.
148 Pfaller AY, Brooks MA, Hetzel S, *et al*. Effect of a New Rule Limiting Full Contact Practice on the Incidence of Sport-Related Concussion in High School Football Players. *Am J Sports Med* 2019;47:2294–9.
149 Bretzin AC, Tomczyk CP, Wiebe DJ, *et al*. Avenues for Sport-Related Concussion Prevention in High School Football: Effect of Limiting Collision Practices. *J Athl Train* 2022;57:733–40.
150 Dick R, Ferrara MS, Agel J, *et al*. Descriptive epidemiology of collegiate men's football injuries: National Collegiate Athletic Association Injury Surveillance System, 1988-1989 through 2003-2004. *J Athl Train* 2007;42:221–33.
151 Stemper BD, Shah AS, Harezlak J, *et al*. Repetitive Head Impact Exposure in College Football Following an NCAA Rule Change to Eliminate Two-A-Day Preseason Practices: A Study from the NCAA-DoD CARE Consortium. *Ann Biomed Eng* 2019;47:2073–85.
152 Stemper BD, Shah AS, Mihalik JP, *et al*. Head Impact Exposure in College Football after a Reduction in Preseason Practices. *Med Sci Sports Exerc* 2020;52:1629–38.
153 Recker R, Myers A, Desai N, *et al*. Headgear use in girls' lacrosse-stakeholders not ready for change. *Front Sports Act Living* 2024;6:1363007.
154 Makovec Knight J, Nguyen JVK, Mitra B, *et al*. Soft-shell headgear, concussion and injury prevention in youth team collision sports: a systematic review. *BMJ Open* 2021;11:e044320.
155 McIntosh AS, McCrory P, Finch CF, *et al*. Does padded headgear prevent head injury in rugby union football? *Med Sci Sports Exerc* 2009;41:306–13.
156 Stokes KA, Cross M, Williams S, *et al*. Padded Headgear does not Reduce the Incidence of Match Concussions in Professional Men's Rugby Union: A Case-control Study of 417 Cases. *Int J Sports Med* 2021;42:930–5.
157 Knapik JJ, Hoedebecke BL, Rogers GG, *et al*. Effectiveness of Mouthguards for the Prevention of Orofacial Injuries and Concussions in Sports: Systematic Review and Meta-Analysis. *Sports Med* 2019;49:1217–32.
158 Stenger JM, Lawson EA, Wright JM, *et al*. Mouthguards: protection against shock. *The Journal of the American Dental Association* 1964;69:273–81.
159 Tanaka Y, Tsugawa T, Maeda Y. Effect of mouthguards on impact to the craniomandibular complex. *Dent Traumatol* 2017;33:51–6.
160 Benson BW, Hamilton GM, Meeuwisse WH, *et al*. Is protective equipment useful in preventing concussion? A systematic review of the literature. *Br J Sports Med* 2009;43 Suppl 1:i56–67.
161 Schneider KJ, Meeuwisse WH, Palacios-Derflingher L, *et al*. Changes in Measures of Cervical Spine Function, Vestibulo-ocular Reflex, Dynamic Balance, and Divided Attention Following Sport-Related Concussion in Elite Youth Ice Hockey Players. *J Orthop Sports Phys Ther* 2018;48:974–81.
162 Collins CL, Fletcher EN, Fields SK, *et al*. Neck Strength: A Protective Factor Reducing Risk for Concussion in High School Sports. *J Primary Prevent* 2014;35:309–19.